In [4]:
```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
```

In [6]:
```
df =pd.read_csv(r'C:\DataScienceAndAICourse\february\28th - Seaborn, Eda Practicle\EDA- HEALTHCARE DOMAIN\heart.csv')
```

In [8]:
```
df.head()
```

Out[8]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

In [12]:
```
df.shape
```

Out[12]:
```
(303, 14)
```

In [14]:
```
df.columns
```

Out[14]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

In [16]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    int64
 1   sex       303 non-null    int64
 2   cp        303 non-null    int64
 3   trestbps  303 non-null    int64
 4   chol      303 non-null    int64
 5   fbs       303 non-null    int64
 6   restecg   303 non-null    int64
 7   thalach   303 non-null    int64
 8   exang     303 non-null    int64
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64
 11  ca        303 non-null    int64
 12  thal      303 non-null    int64
 13  target    303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [20]:
```
df.dtypes
```

Out[20]:
```
age           int64
sex           int64
cp            int64
trestbps      int64
chol          int64
fbs           int64
restecg       int64
thalach       int64
exang         int64
oldpeak     float64
slope         int64
ca            int64
thal          int64
target        int64
dtype: object
```

In [30]:
```
df.isnull().sum().sum()
```

Out[30]: 0

In [40]:
```
df.notnull().sum().sum()
```

Out[40]: 4242

In [42]:
```
df.size
```

Out[42]: 4242

In [44]:
```
df.describe()
```

Out[44]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733	1.039604
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794	1.161075
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000	0.800000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000	1.600000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000

In [54]:
```
df.describe(include=['int64'])
```

Out[54]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	slope
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733	1.399340
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794	0.616226
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	1.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000	2.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	2.000000

In [58]:

```
df.describe(include='all')
```

Out[58]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733	1.039604
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794	1.161075
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000	0.800000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000	1.600000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000

In [60]:

```
df['target'].unique()
```

Out[60]:
```
array([1, 0], dtype=int64)
```

In [62]:
```
df['target'].nunique()
```

Out[62]:
```
2
```

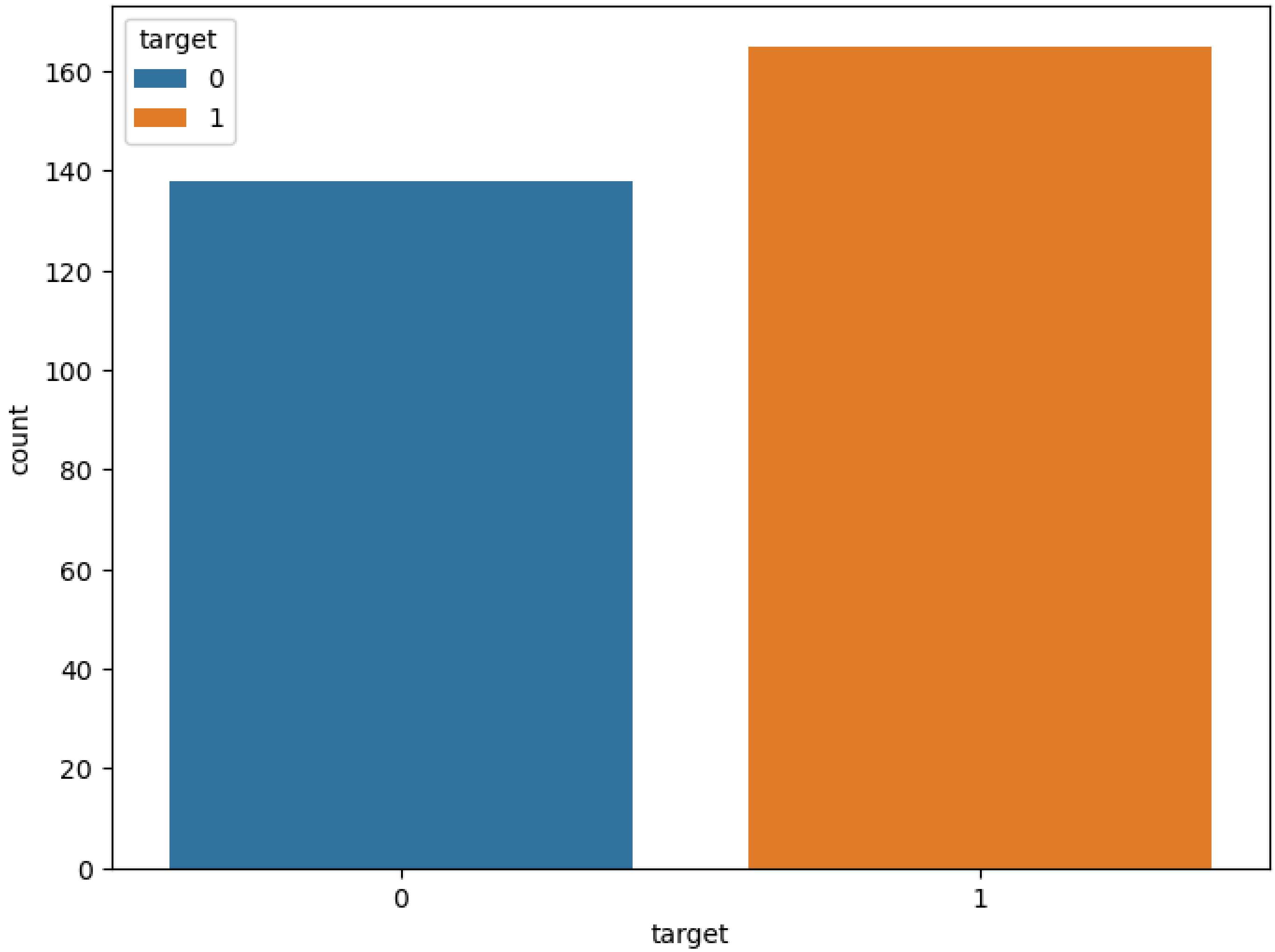
In [64]:
```
df.groupby('target')
```

Out[64]:
```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x000002446ED66C90>
```

In [70]:
```
df['target'].value_counts()
```

Out[70]:
```
target
1    165
0    138
Name: count, dtype: int64
```

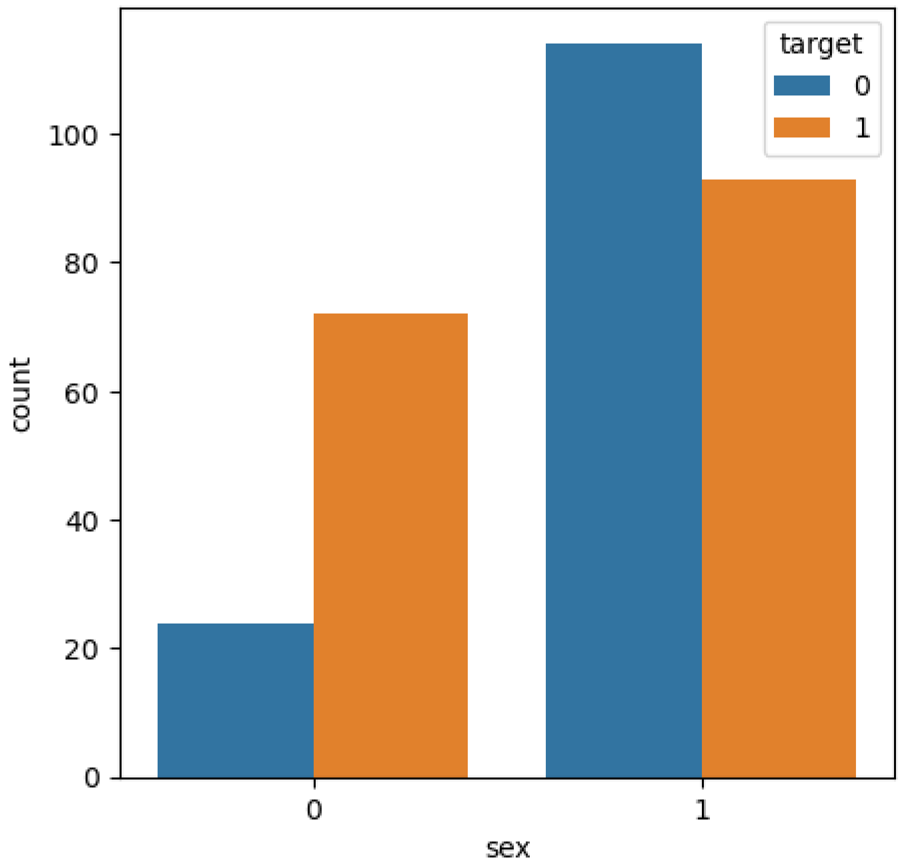
In [76]:
```
f,ax= plt.subplots(figsize=(8,6))
ax= sns.countplot(x='target',hue='target',data=df)
plt.show()
```



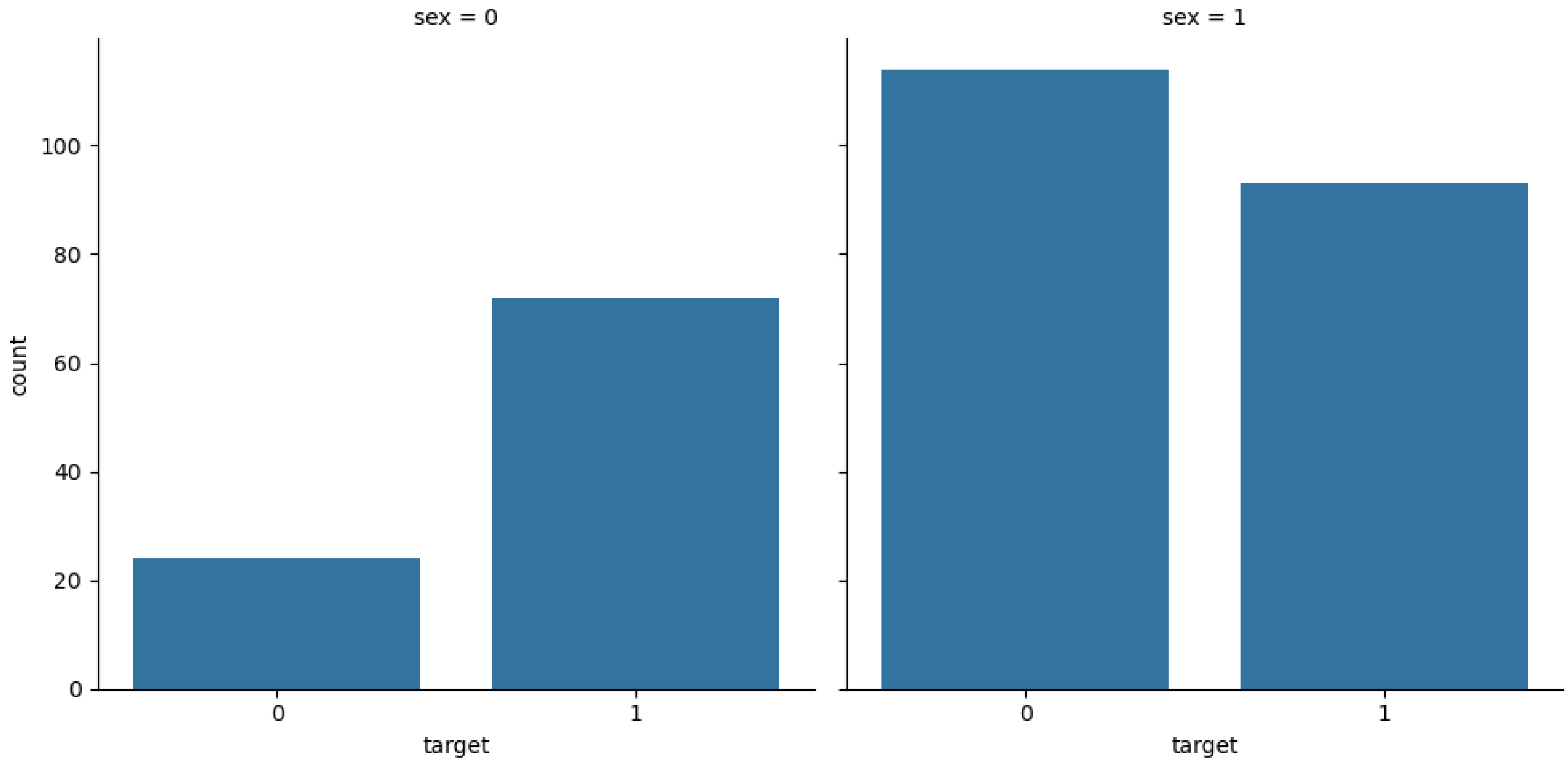
In [80]:
```
df.groupby('sex')['target'].value_counts()
```

Out[80]:
```
sex  target
0    1          72
     0          24
1    0         114
     1          93
Name: count, dtype: int64
```

In [84]:
```
f,ax=plt.subplots(figsize=(5,5))
ax = sns.countplot(x='sex',hue='target',data=df)
```

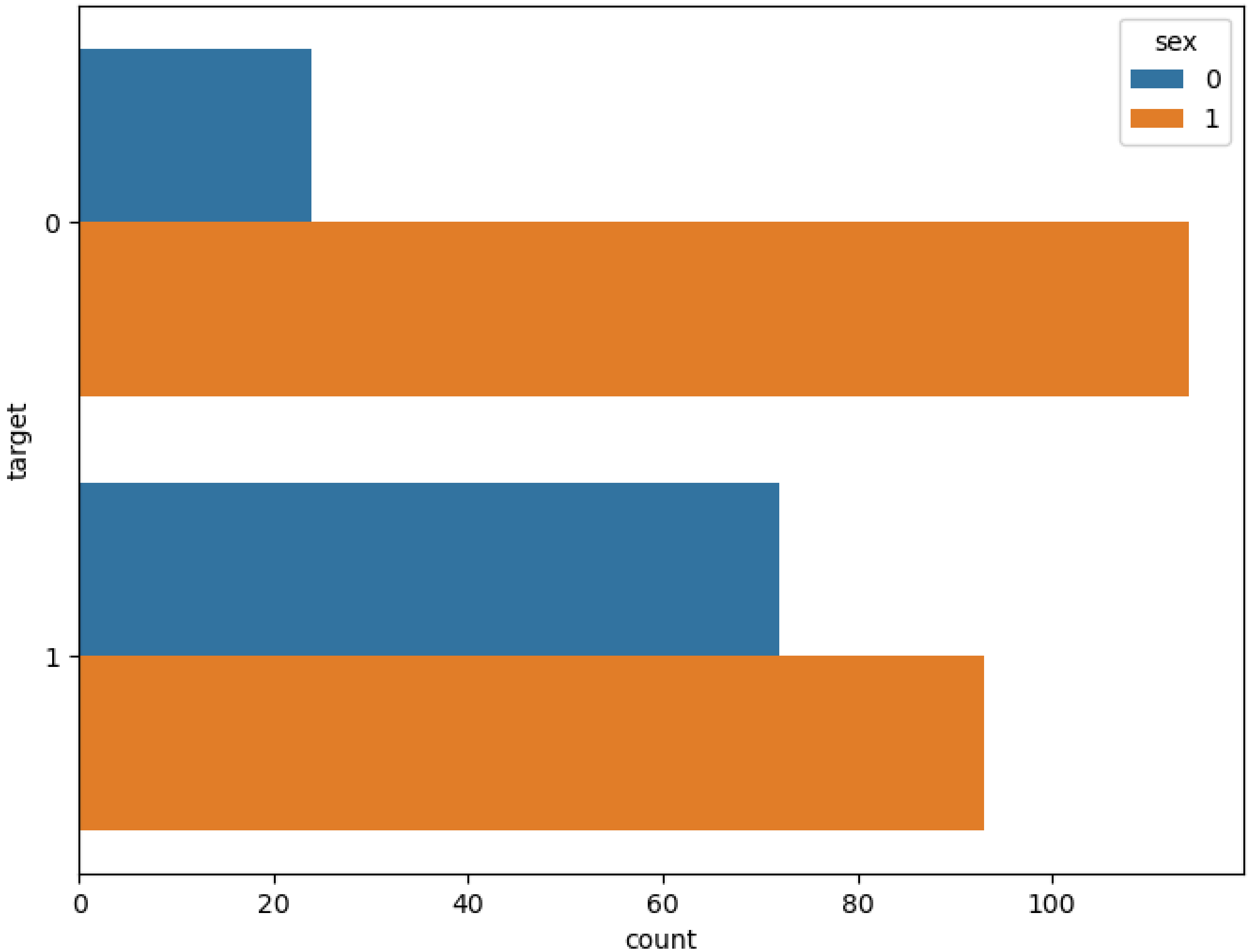


In [94]:
```
ax = sns.catplot(x='target',col='sex' , kind='count',data=df)
```



In [104…

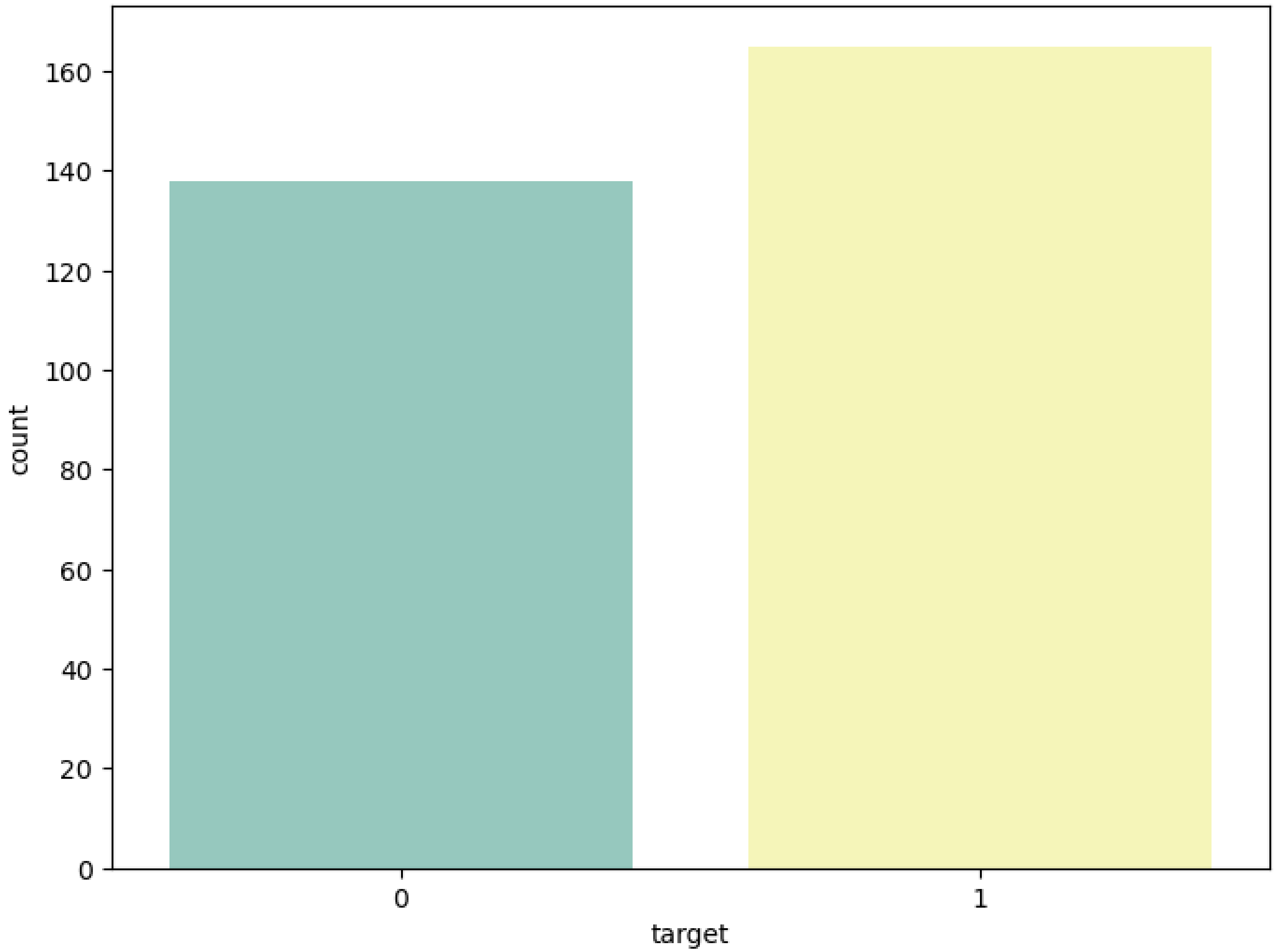
```
f,ax= plt.subplots(figsize=(8,6))
ax= sns.countplot(y='target',hue='sex',data=df)
```



In [110…

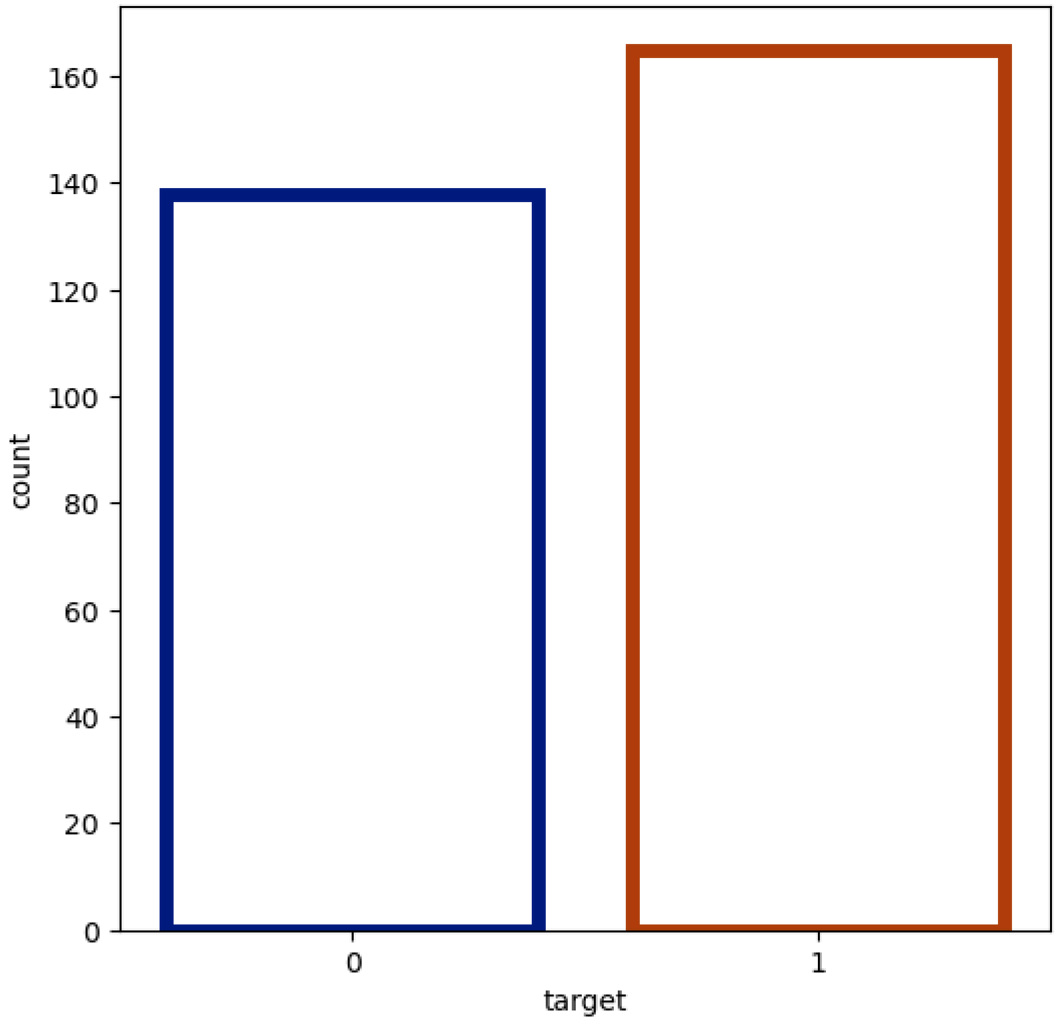
```
import warnings
warnings.filterwarnings('ignore')

f,ax= plt.subplots(figsize=(8,6))
ax= sns.countplot(x='target',data=df,palette='Set3')
```

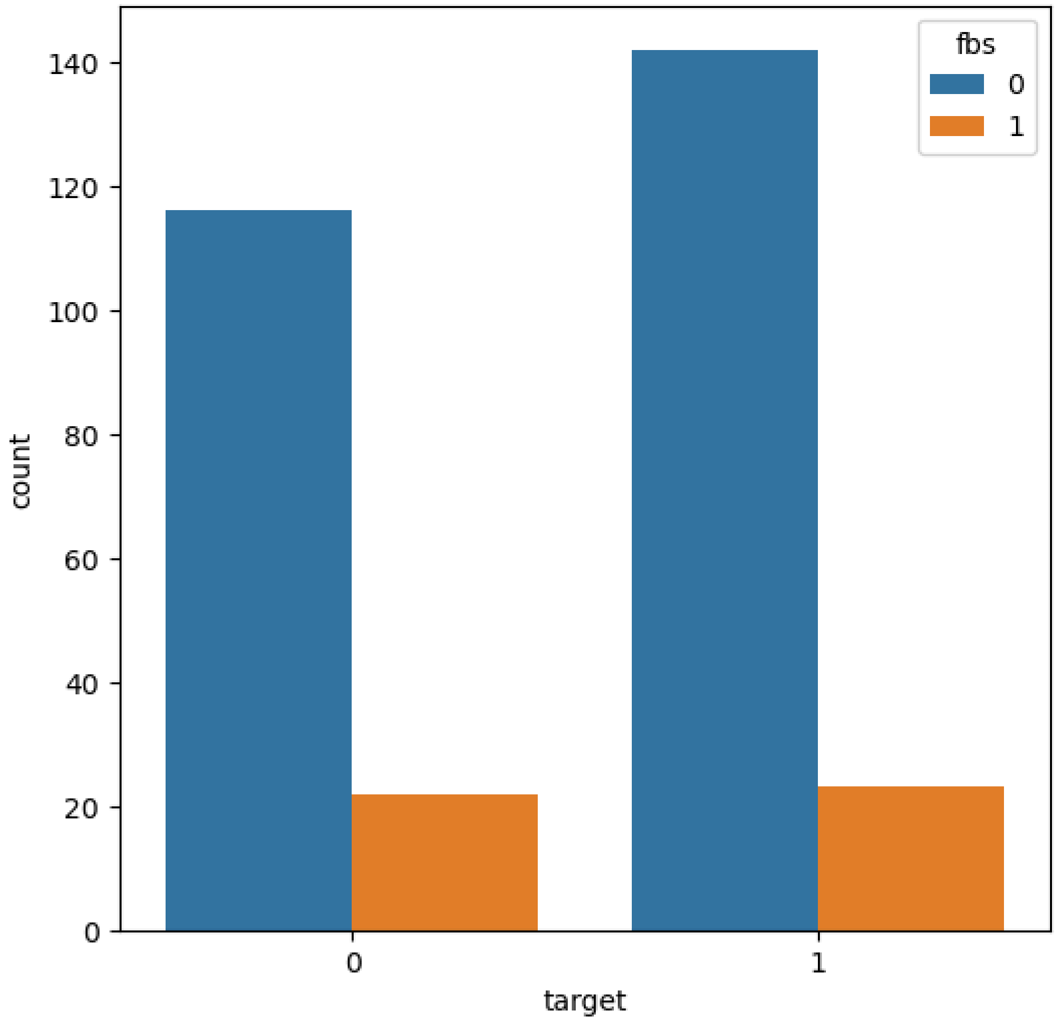


In [130…

```
f,ax= plt.subplots(figsize=(6,6))
ax = sns.countplot(x='target',data=df,facecolor=(0,0,0,0),linewidth=5,edgecolor=sns.color_palette("dark", 3))
```

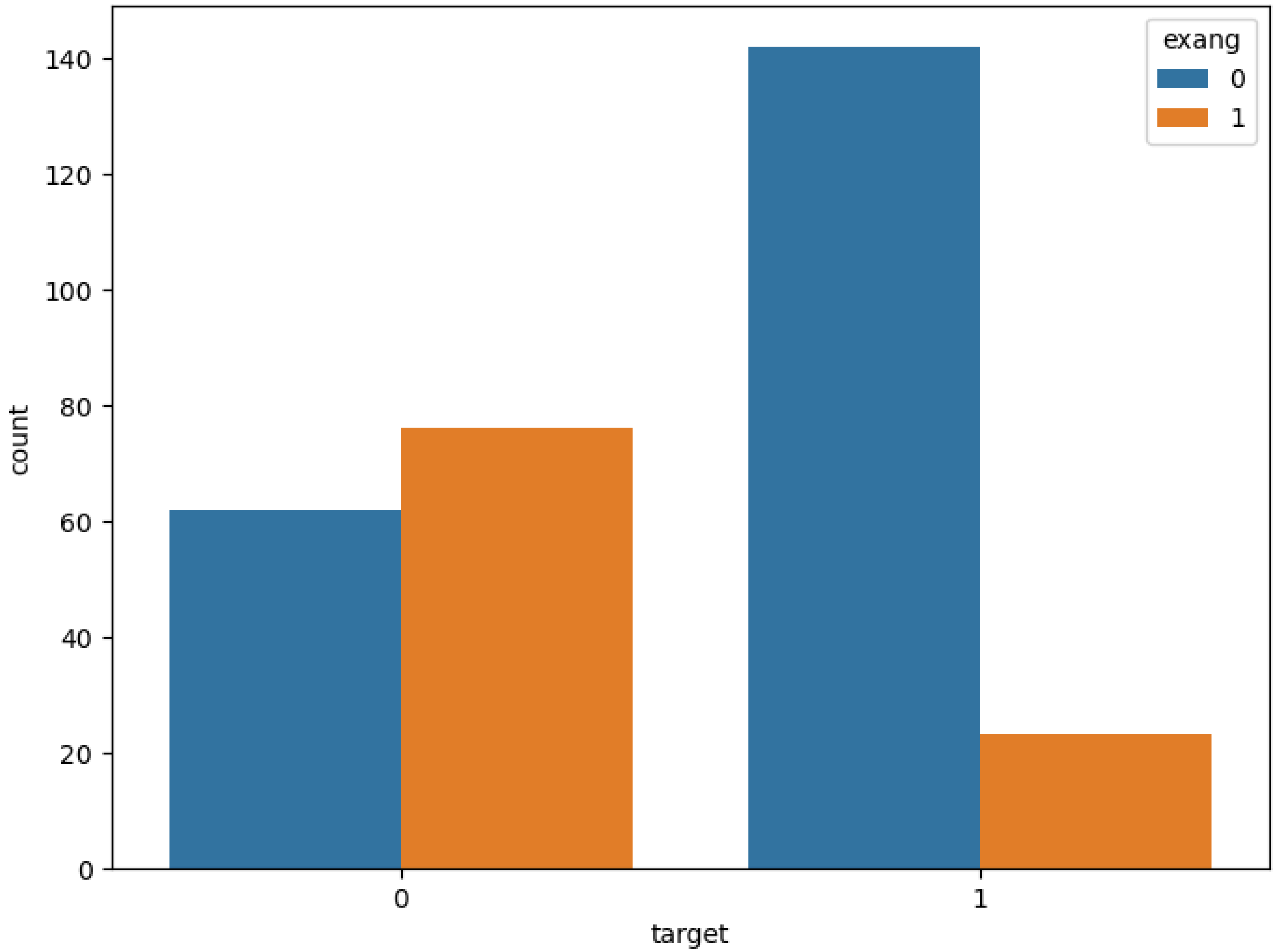


In [134…
```
f,ax=plt.subplots(figsize=(6,6))
ax = sns.countplot(x='target',hue='fbs',data=df)
```



In [136…

```
f,ax= plt.subplots(figsize=(8,6))
ax= sns.countplot(x='target',hue='exang',data=df)
```



In [138…

```
correlation = df.corr()
```

In [140…

```
correlation
```

Out[140…

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
age	1.000000	-0.098447	-0.068653	0.279351	0.213678	0.121308	-0.116211	-0.398522	0.096801	0.210013	-0.168814
sex	-0.098447	1.000000	-0.049353	-0.056769	-0.197912	0.045032	-0.058196	-0.044020	0.141664	0.096093	-0.030711
cp	-0.068653	-0.049353	1.000000	0.047608	-0.076904	0.094444	0.044421	0.295762	-0.394280	-0.149230	0.119717
trestbps	0.279351	-0.056769	0.047608	1.000000	0.123174	0.177531	-0.114103	-0.046698	0.067616	0.193216	-0.121475
chol	0.213678	-0.197912	-0.076904	0.123174	1.000000	0.013294	-0.151040	-0.009940	0.067023	0.053952	-0.004038
fbs	0.121308	0.045032	0.094444	0.177531	0.013294	1.000000	-0.084189	-0.008567	0.025665	0.005747	-0.059894
restecg	-0.116211	-0.058196	0.044421	-0.114103	-0.151040	-0.084189	1.000000	0.044123	-0.070733	-0.058770	0.093045
thalach	-0.398522	-0.044020	0.295762	-0.046698	-0.009940	-0.008567	0.044123	1.000000	-0.378812	-0.344187	0.386784
exang	0.096801	0.141664	-0.394280	0.067616	0.067023	0.025665	-0.070733	-0.378812	1.000000	0.288223	-0.257748
oldpeak	0.210013	0.096093	-0.149230	0.193216	0.053952	0.005747	-0.058770	-0.344187	0.288223	1.000000	-0.577537
slope	-0.168814	-0.030711	0.119717	-0.121475	-0.004038	-0.059894	0.093045	0.386784	-0.257748	-0.577537	1.000000
ca	0.276326	0.118261	-0.181053	0.101389	0.070511	0.137979	-0.072042	-0.213177	0.115739	0.222682	-0.080155
thal	0.068001	0.210041	-0.161736	0.062210	0.098803	-0.032019	-0.011981	-0.096439	0.206754	0.210244	-0.104764
target	-0.225439	-0.280937	0.433798	-0.144931	-0.085239	-0.028046	0.137230	0.421741	-0.436757	-0.430696	0.345877

In [142…

```
correlation['target']
```

Out[142…

```
age        -0.225439
sex        -0.280937
cp          0.433798
trestbps   -0.144931
chol       -0.085239
fbs        -0.028046
restecg     0.137230
thalach     0.421741
exang      -0.436757
oldpeak    -0.430696
slope       0.345877
ca         -0.391724
thal       -0.344029
target      1.000000
Name: target, dtype: float64
```

In [146…

```
correlation['target'].sort_values(ascending=False)
```

Out[146…

```
target      1.000000
cp          0.433798
thalach     0.421741
slope       0.345877
restecg     0.137230
fbs        -0.028046
chol       -0.085239
trestbps   -0.144931
age        -0.225439
sex        -0.280937
thal       -0.344029
ca         -0.391724
oldpeak    -0.430696
exang      -0.436757
Name: target, dtype: float64
```

In [148…

```
df['cp'].unique()
```

Out[148…

```
array([3, 2, 1, 0], dtype=int64)
```

In [150…

```
df['cp'].nunique()
```

Out[150…

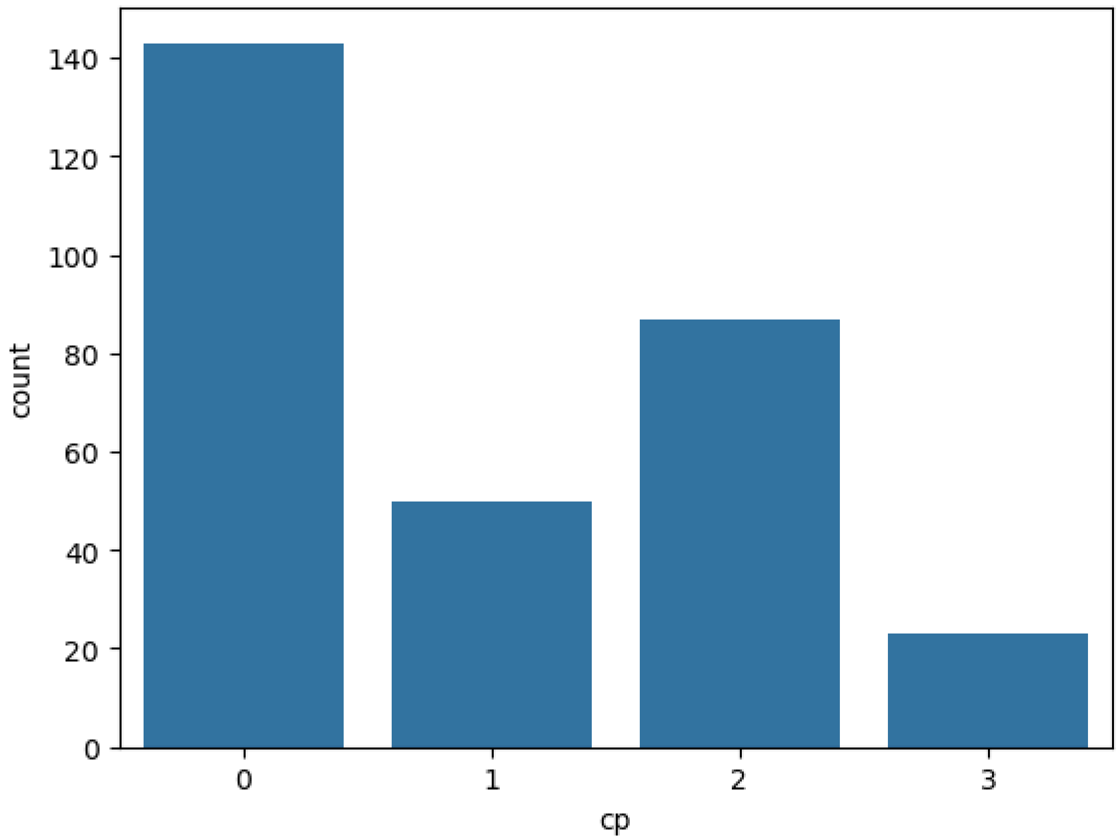
```
4
```

In [152…
```
df['cp'].value_counts()
```

Out[152…
```
cp
0    143
2     87
1     50
3     23
Name: count, dtype: int64
```

In [154…
```
sns.countplot(x='cp',data=df)
```

Out[154…
```
<Axes: xlabel='cp', ylabel='count'>
```



In [156…
```
df.groupby('cp')['target'].value_counts()
```

Out[156…

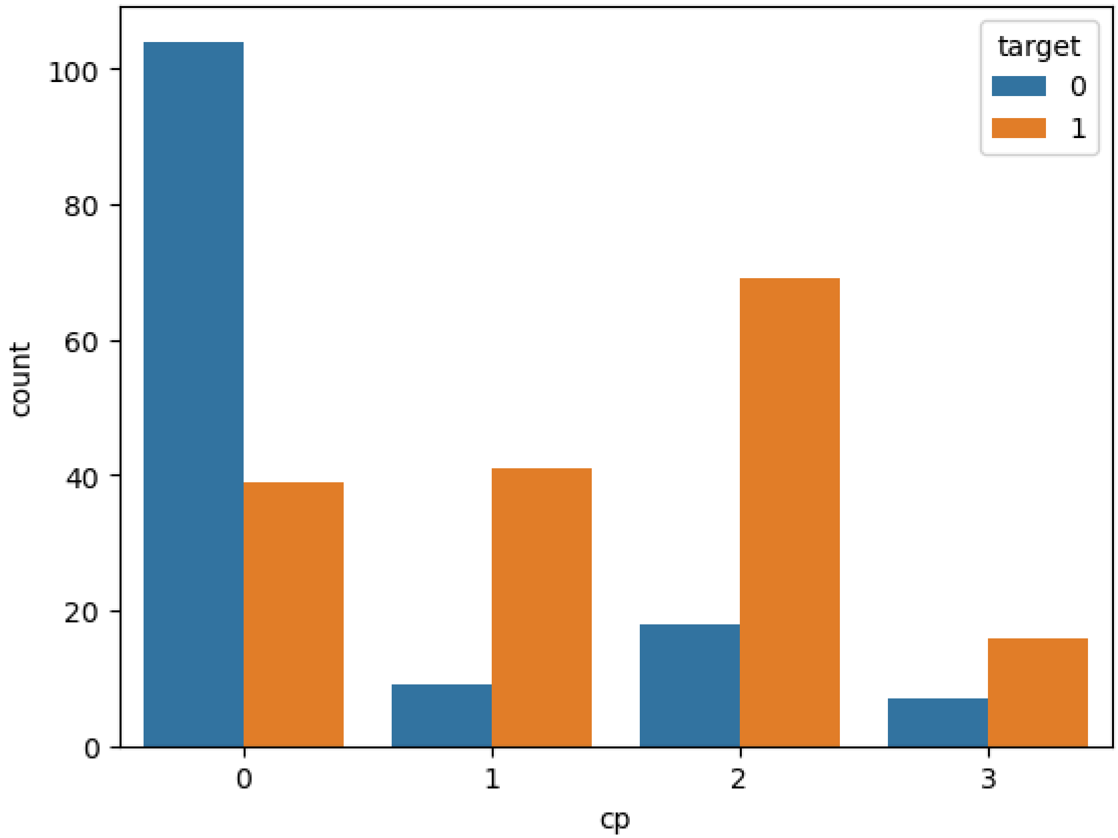
```
cp  target
0   0         104
    1          39
1   1          41
    0           9
2   1          69
    0          18
3   1          16
    0           7
Name: count, dtype: int64
```

In [158…

```
sns.countplot(x='cp',hue='target',data=df)
```

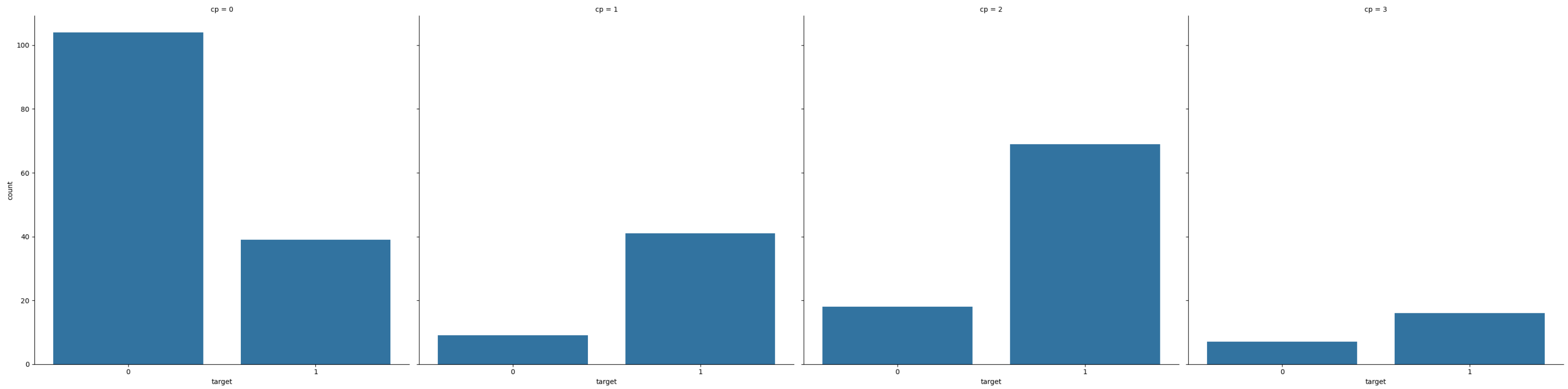
Out[158…

```
<Axes: xlabel='cp', ylabel='count'>
```



In [170…
```
sns.catplot(kind='count',x='target',data=df,col='cp',height=8)
```

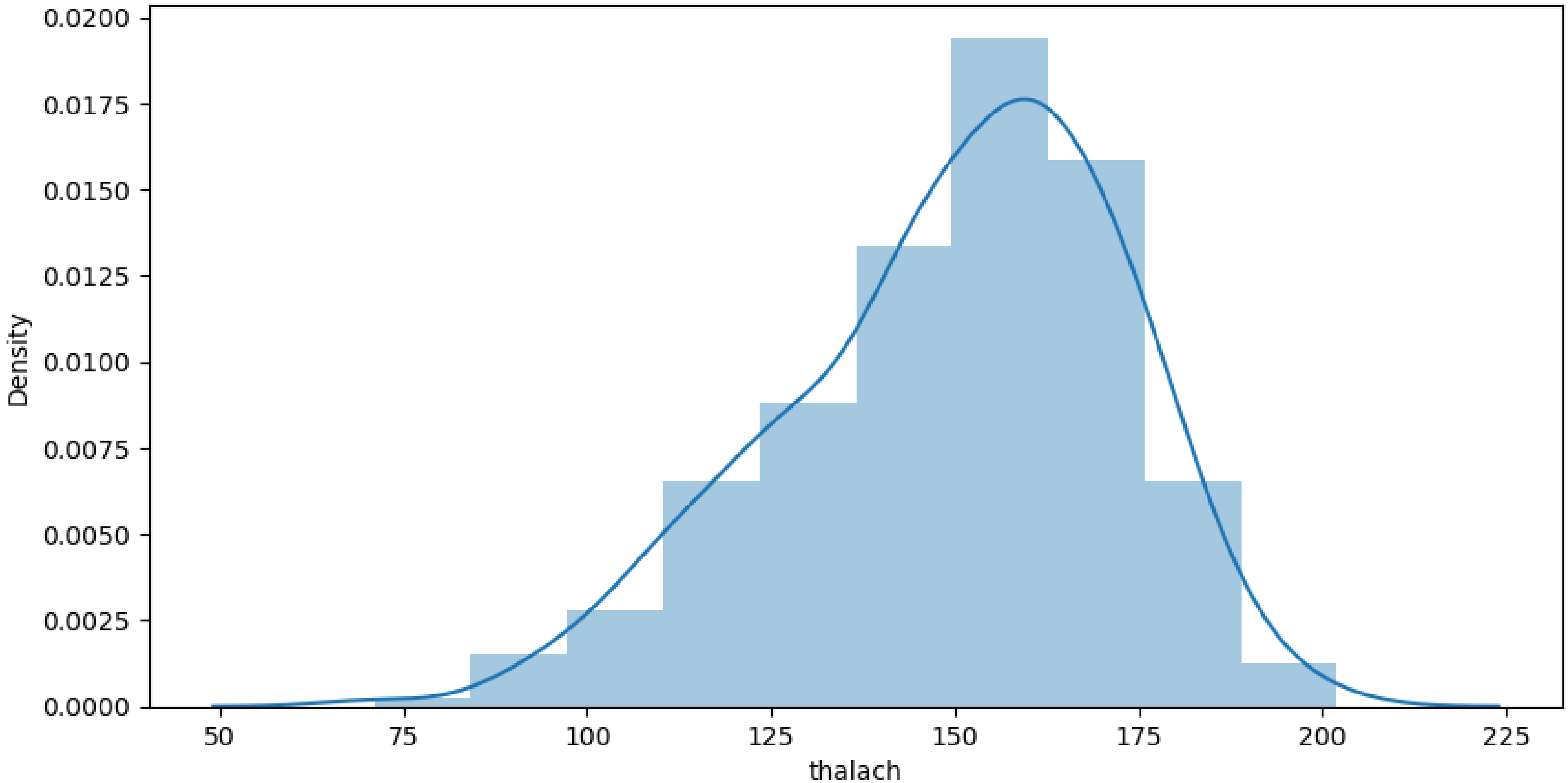
Out[170…
```
<seaborn.axisgrid.FacetGrid at 0x24476344dd0>
```



In [172…
```
df['thalach'].nunique()
```

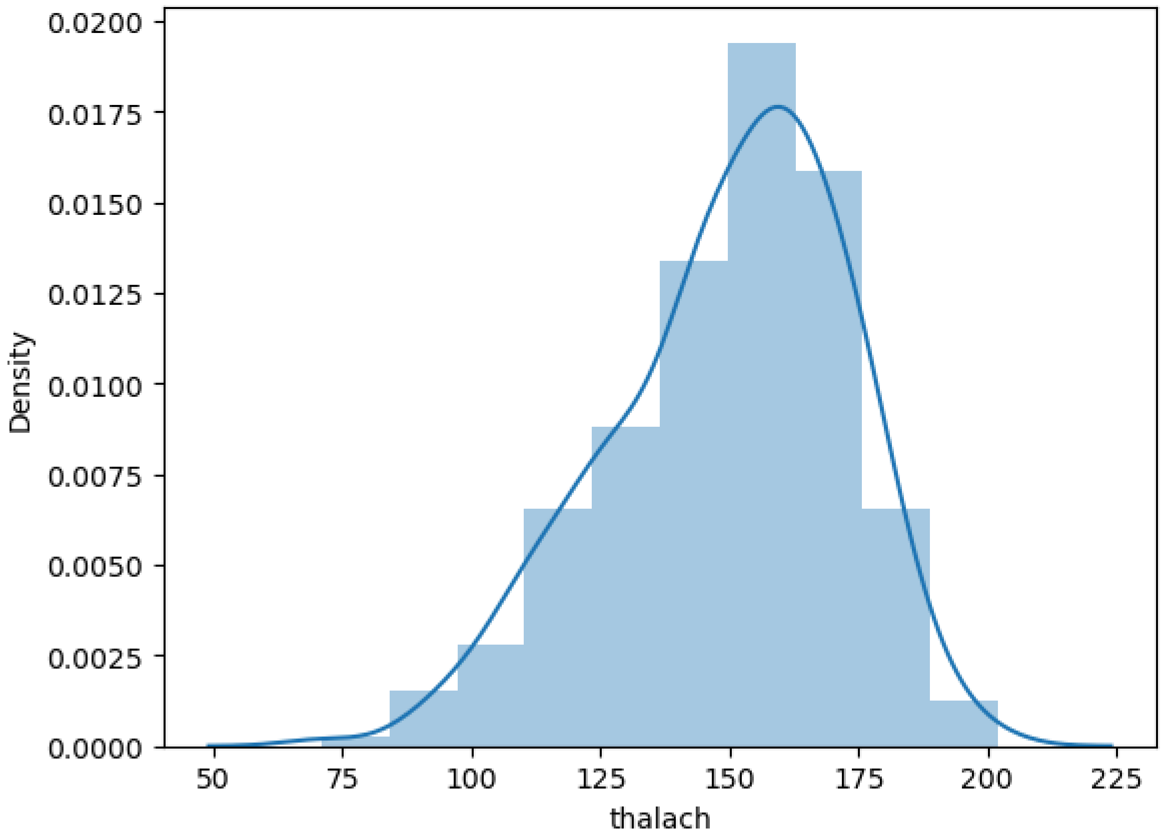
Out[172…
```
91
```

In [180…
```
f,ax= plt.subplots(figsize=(10,5))
x= df['thalach']
ax= sns.distplot(x,bins=10)
```



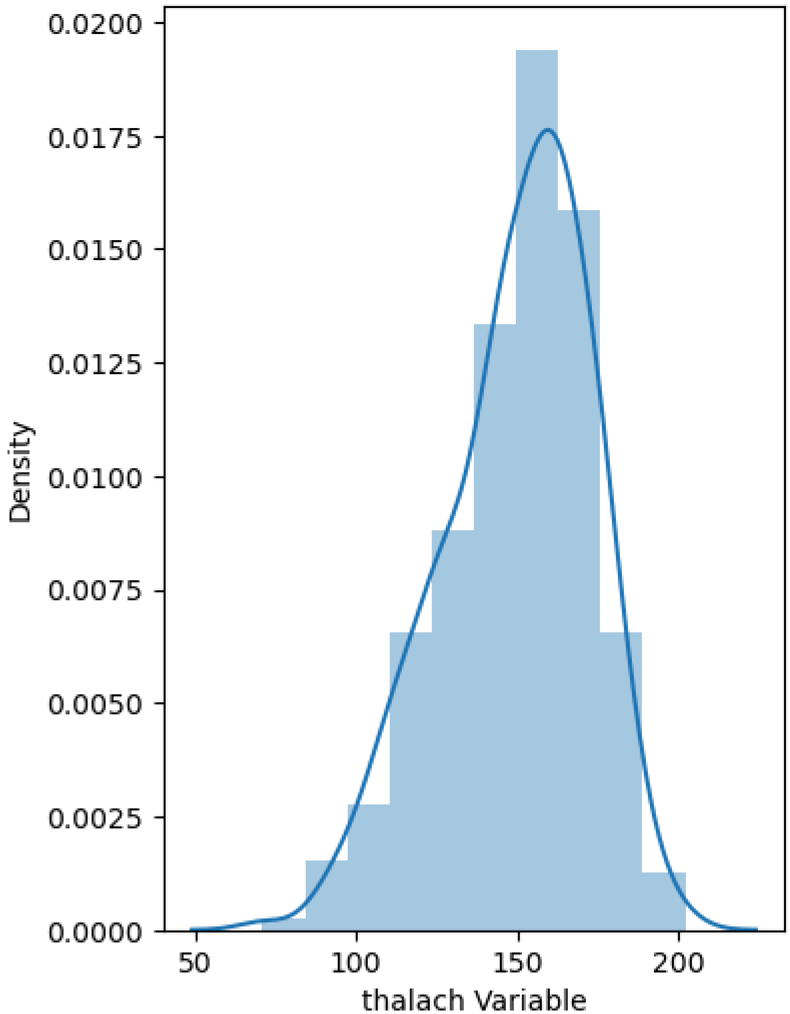
In [184…
```
sns.distplot(df['thalach'],bins=10)
```

Out[184…
```
<Axes: xlabel='thalach', ylabel='Density'>
```



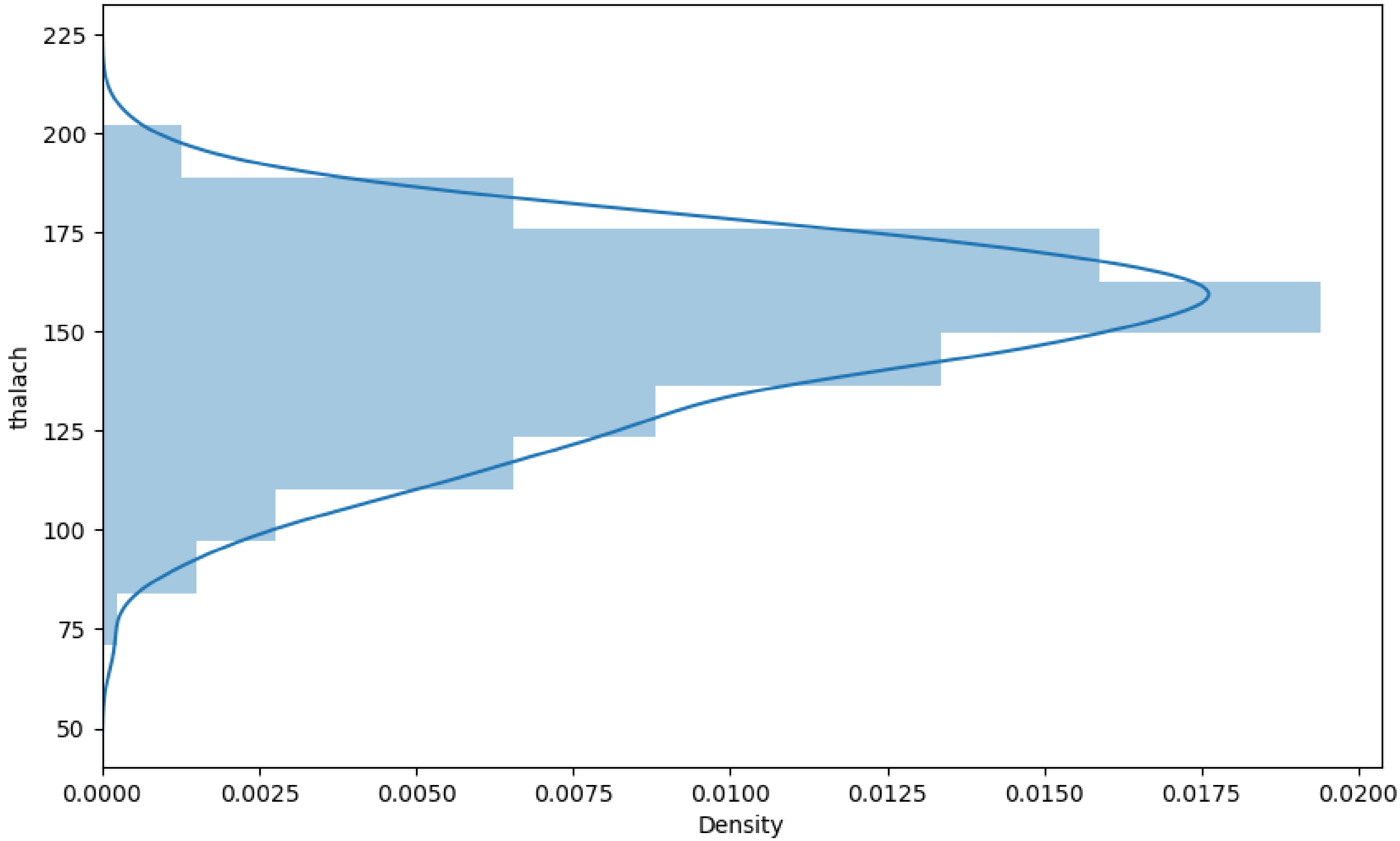
In [188…

```
f,ax= plt.subplots(figsize=(4,6))
x=df['thalach']
x= pd.Series(x, name='thalach Variable')
ax= sns.distplot(x,bins=10)
```



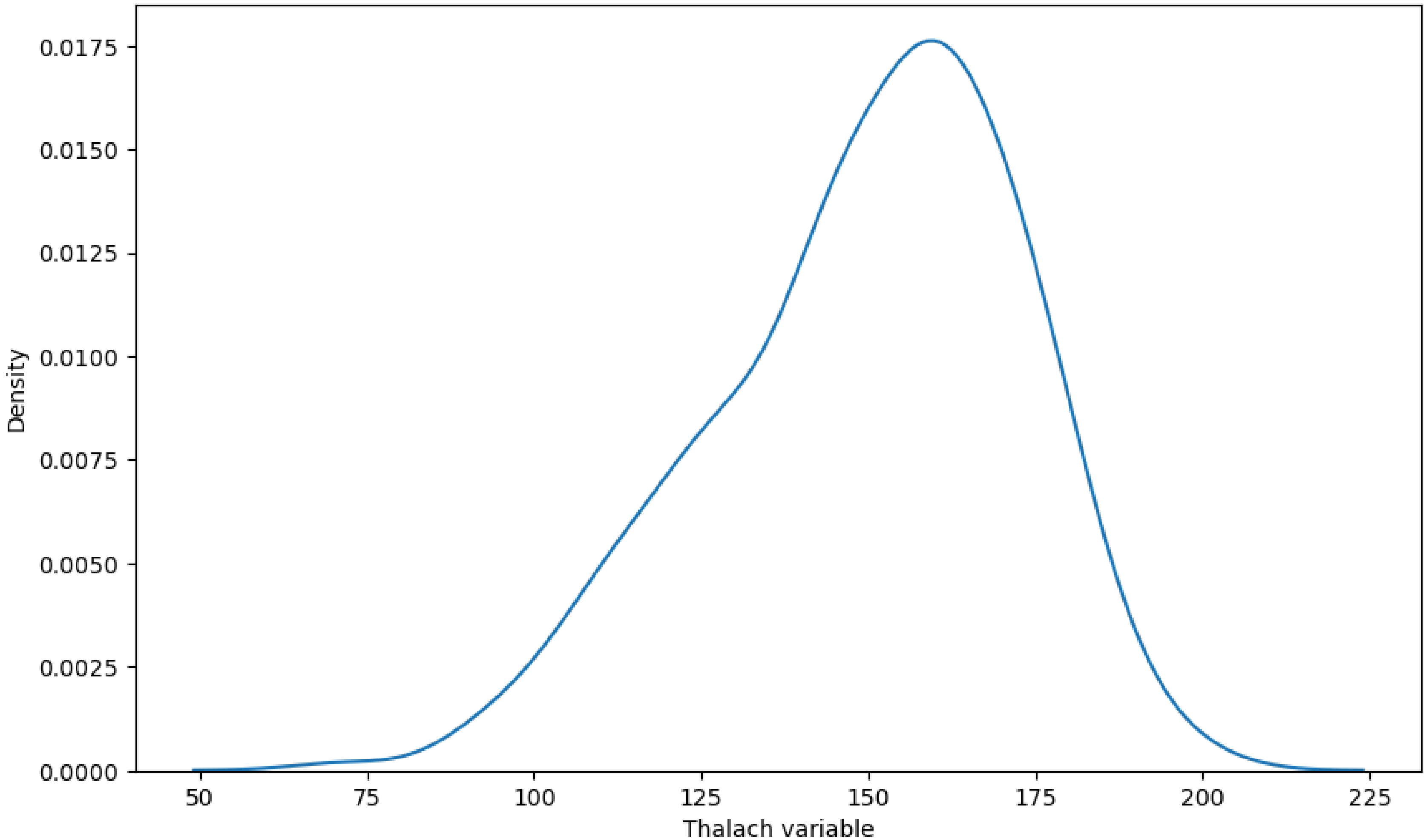
In [196…

```
f,ax = plt.subplots(figsize=(10,6))
x= df['thalach']
ax= sns.distplot(x,bins=10,vertical=True)
```



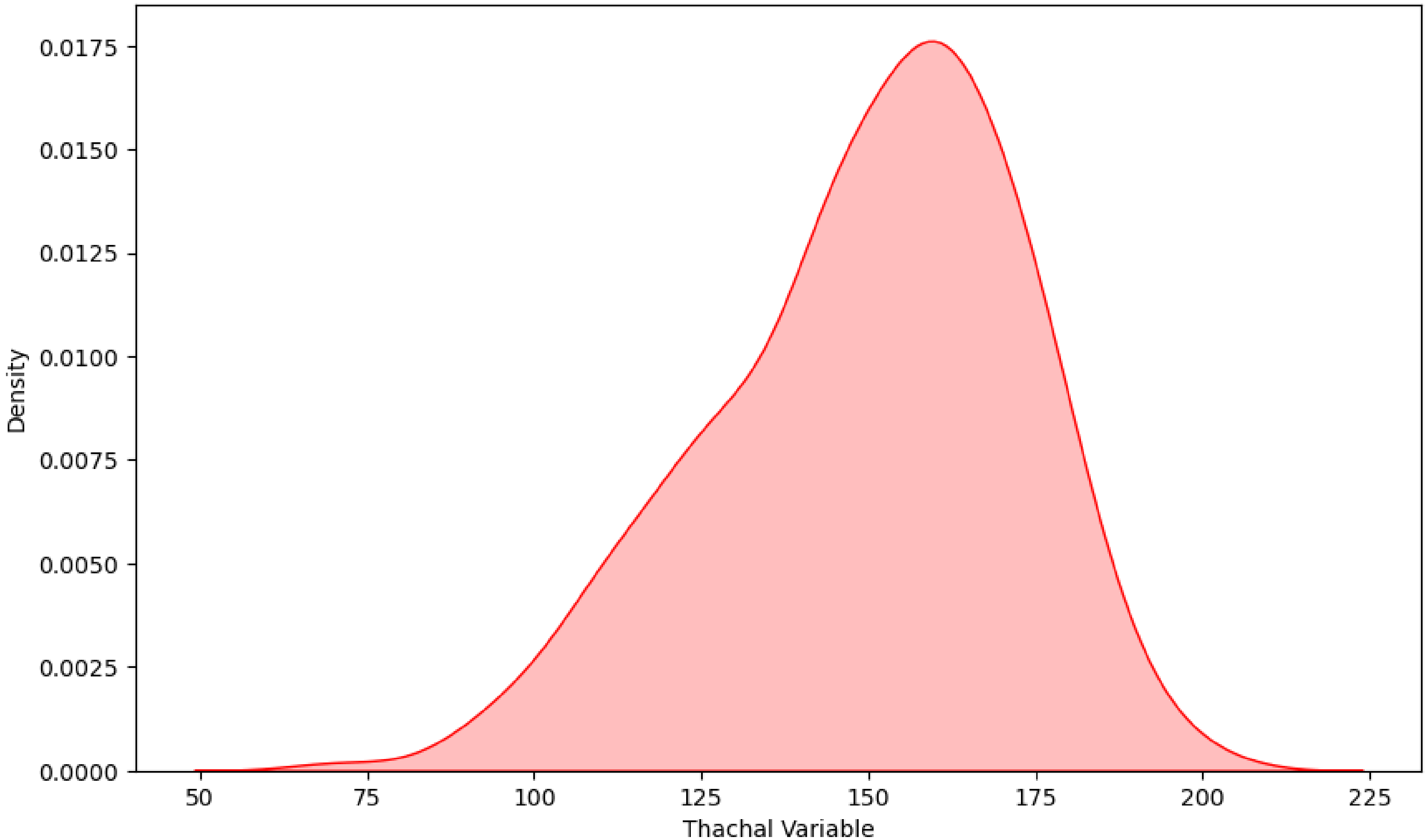
In [198…

```
f,ax= plt.subplots(figsize=(10,6))
x=df['thalach']
x= pd.Series(x,name='Thalach variable')
ax= sns.kdeplot(x)
```



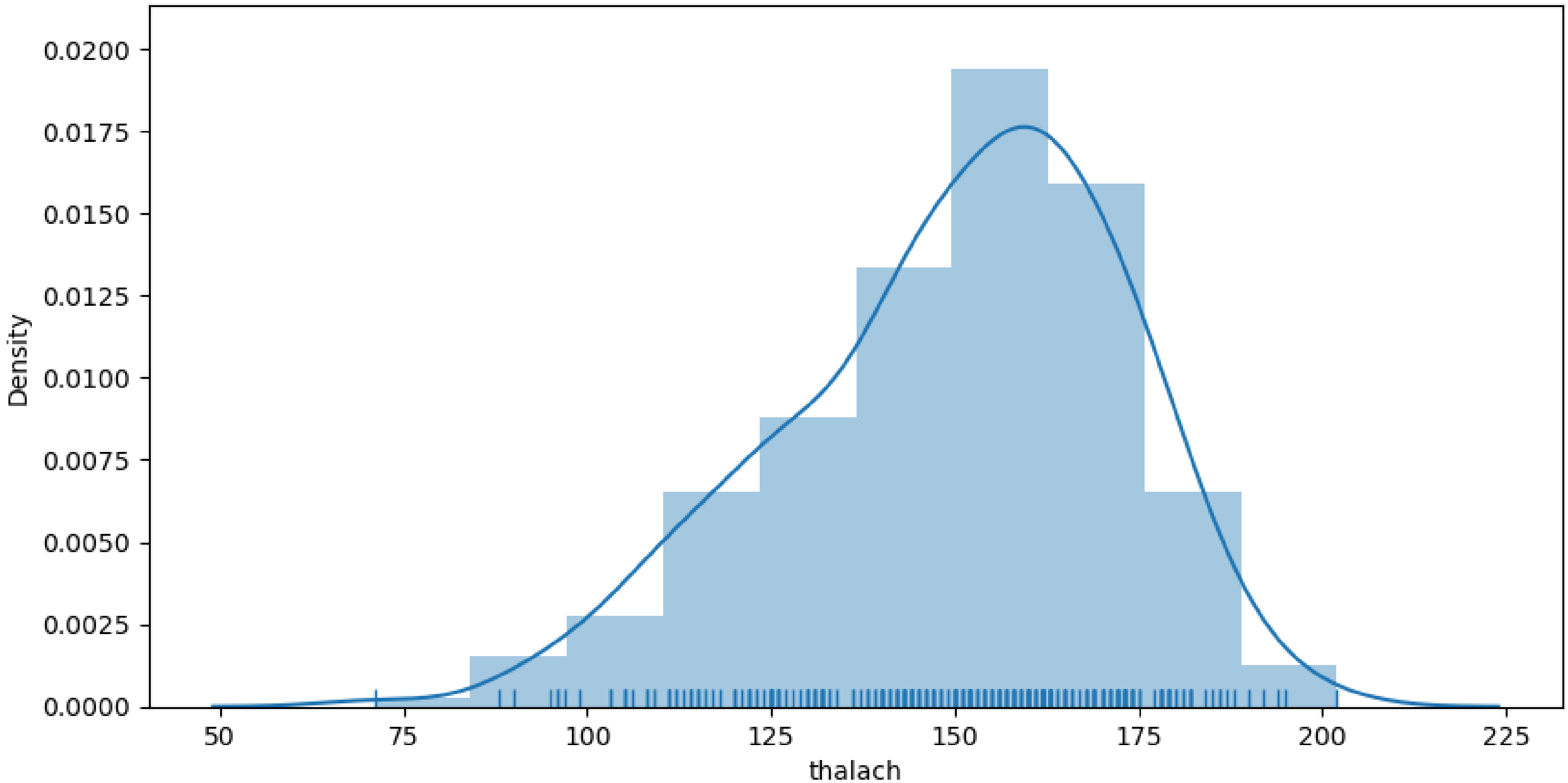
In [204…
```
f,ax= plt.subplots(figsize=(10,6))
x=df['thalach']
x=pd.Series(x,name='Thachal Variable')
sns.kdeplot(x,shade=True,color='r')
```

Out[204…
```
<Axes: xlabel='Thachal Variable', ylabel='Density'>
```



In [214…

```
f,ax = plt.subplots(figsize=(10,5))
x= df['thalach']
ax = sns.distplot(x,kde=True,bins=10,rug=True)
```

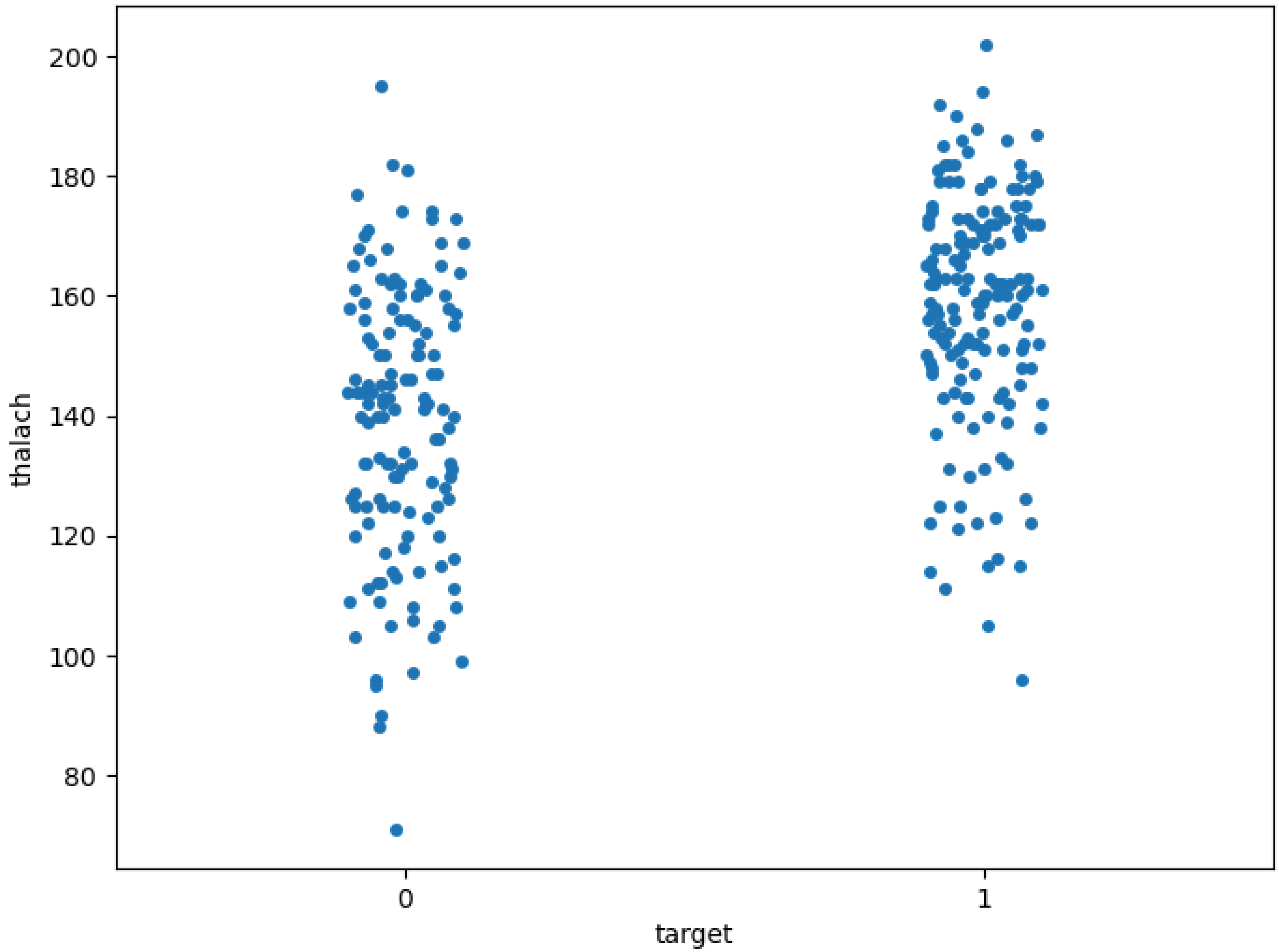


In [216…

```
f,ax = plt.subplots(figsize=(8,6))
sns.stripplot(x='target',y='thalach',data=df)
```

Out[216…

```
<Axes: xlabel='target', ylabel='thalach'>
```

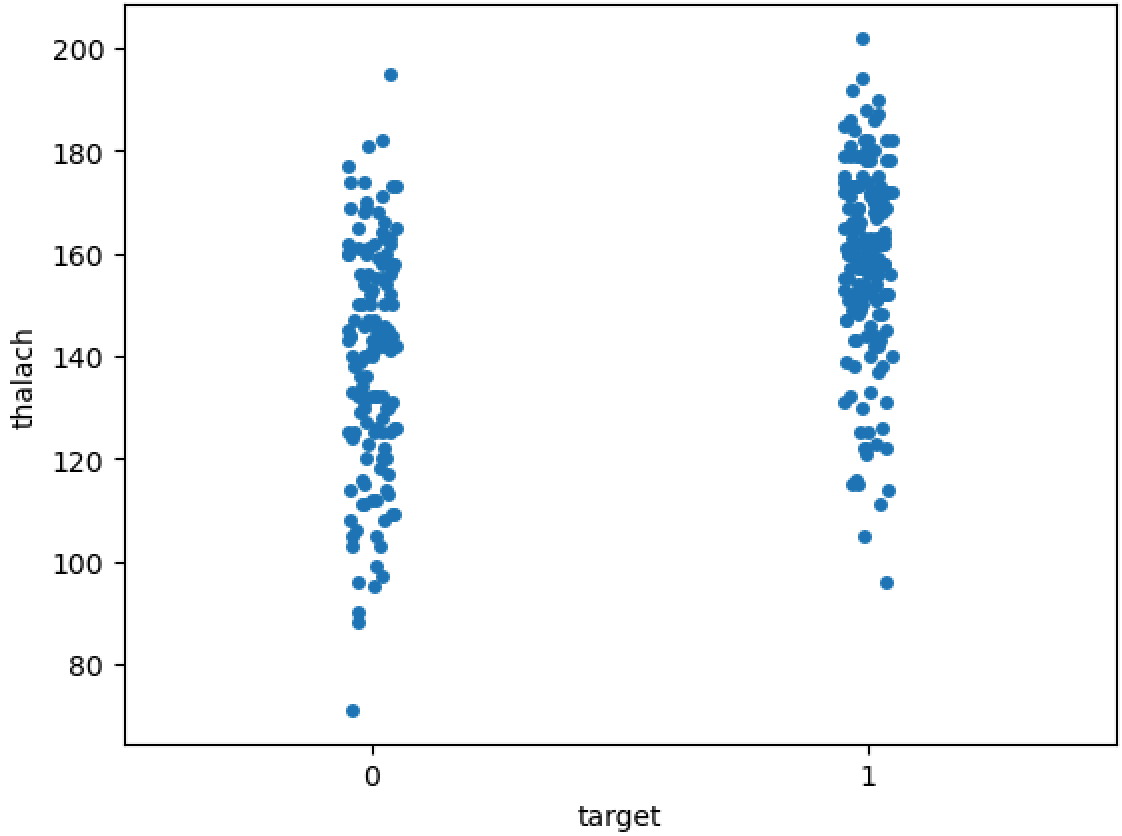


In [230…

```
sns.stripplot(x='target',y='thalach',data=df,jitter=0.05)
```

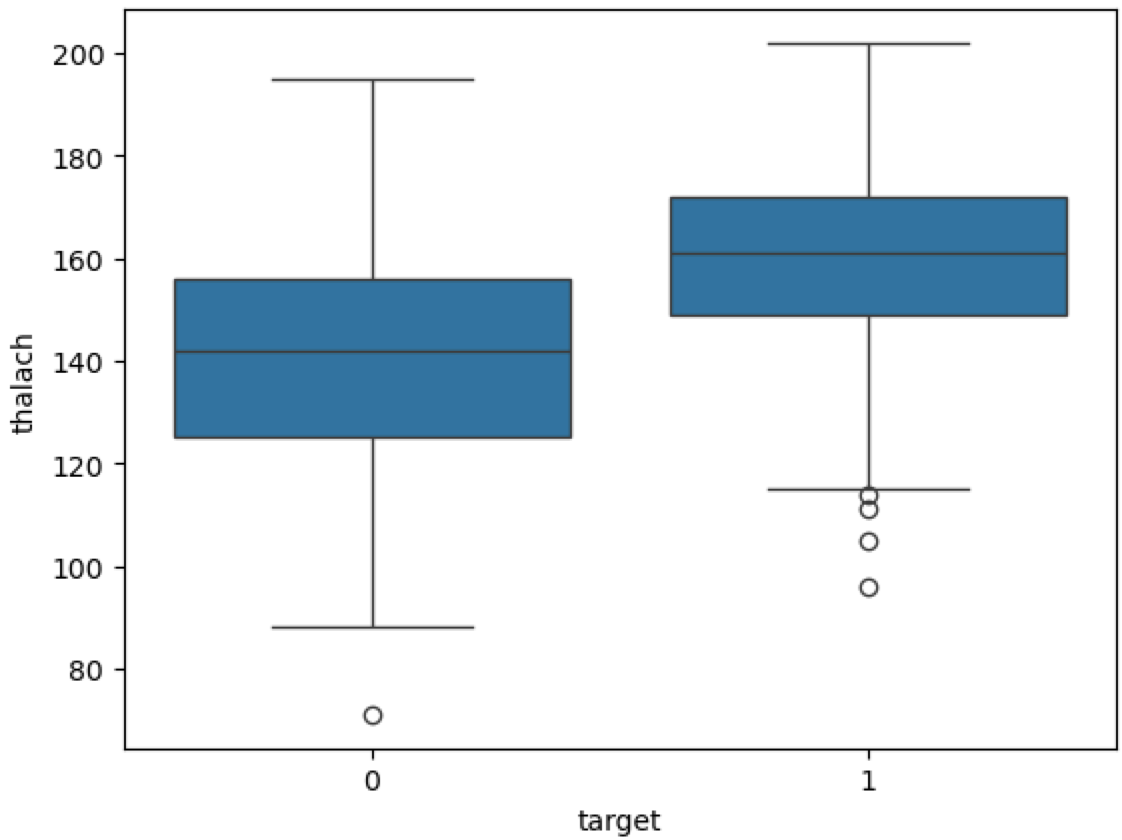
Out[230…

```
<Axes: xlabel='target', ylabel='thalach'>
```



In [232…
```
sns.boxplot(x='target',y='thalach',data=df)
```

Out[232…
```
<Axes: xlabel='target', ylabel='thalach'>
```

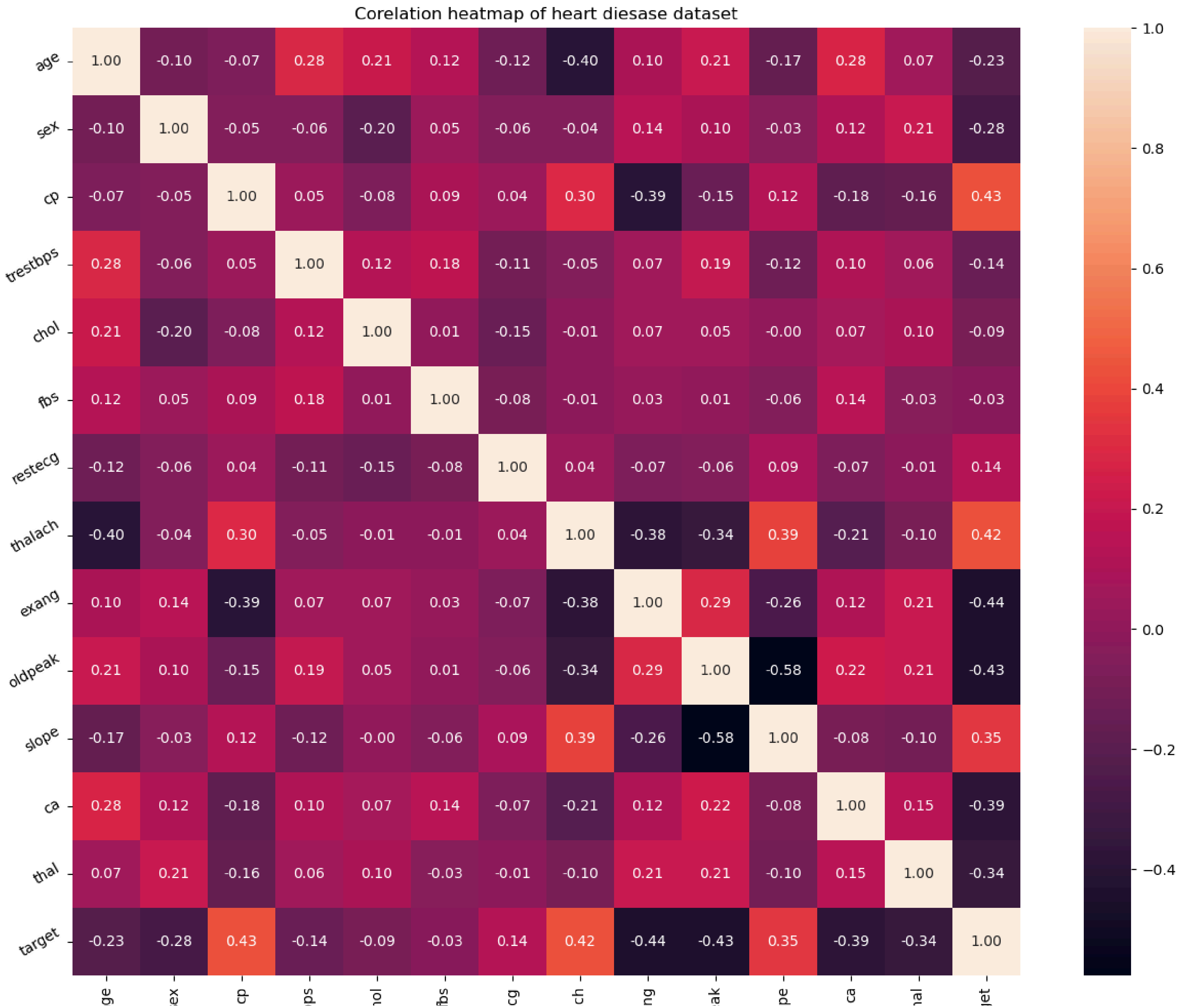


In [246…

```
correlation=df.corr()
plt.figure(figsize=(16,12))
plt.title('Corelation heatmap of heart diesase dataset')
a=sns.heatmap(correlation,square=True,annot=True,fmt='.2f',linecolor='white')
a.set_xticklabels(a.get_xticklabels(),rotation=90)
a.set_yticklabels(a.get_yticklabels(),rotation=30)
```

Out[246…

```
[Text(0, 0.5, 'age'),
 Text(0, 1.5, 'sex'),
 Text(0, 2.5, 'cp'),
 Text(0, 3.5, 'trestbps'),
 Text(0, 4.5, 'chol'),
 Text(0, 5.5, 'fbs'),
 Text(0, 6.5, 'restecg'),
 Text(0, 7.5, 'thalach'),
 Text(0, 8.5, 'exang'),
 Text(0, 9.5, 'oldpeak'),
 Text(0, 10.5, 'slope'),
 Text(0, 11.5, 'ca'),
 Text(0, 12.5, 'thal'),
 Text(0, 13.5, 'target')]
```

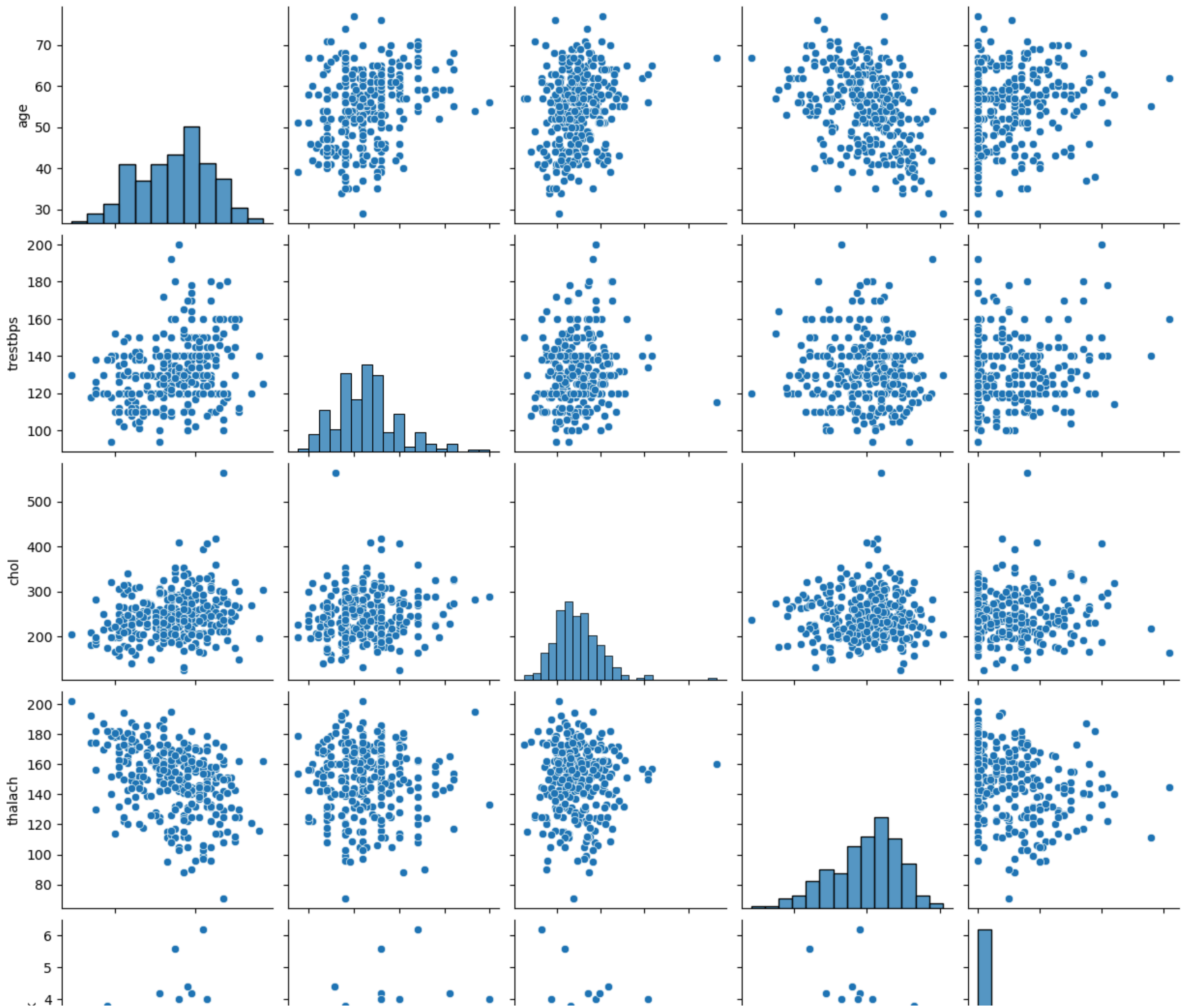

Corelation heatmap of heart diesase dataset
age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

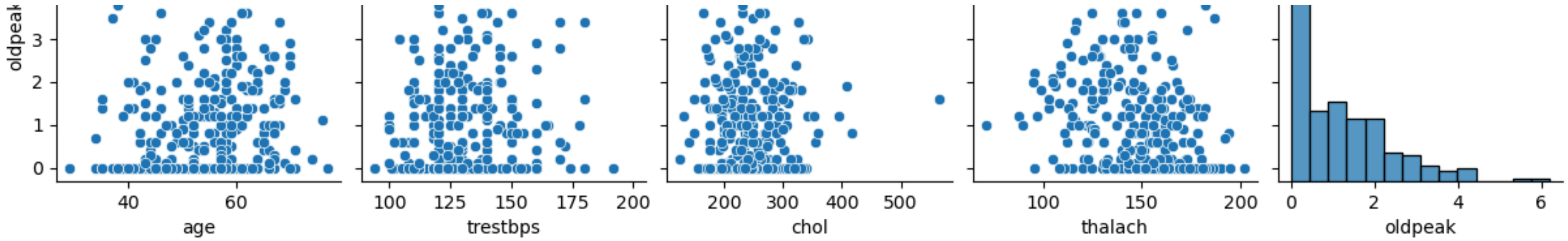
In [256…

```
num_var= ['age','trestbps','chol','thalach','oldpeak']
sns.pairplot(df[num_var],kind='scatter',diag_kind='hist')
```

Out[256…

```
<seaborn.axisgrid.PairGrid at 0x244047ca660>
```


age
trestbps
chol
thalach
70
60
50
40
30
200
180
160
140
120
100
500
400
300
200
80
6
5
4



In [258…

```
df['age'].nunique()
```

Out[258…

```
41
```

In [260…

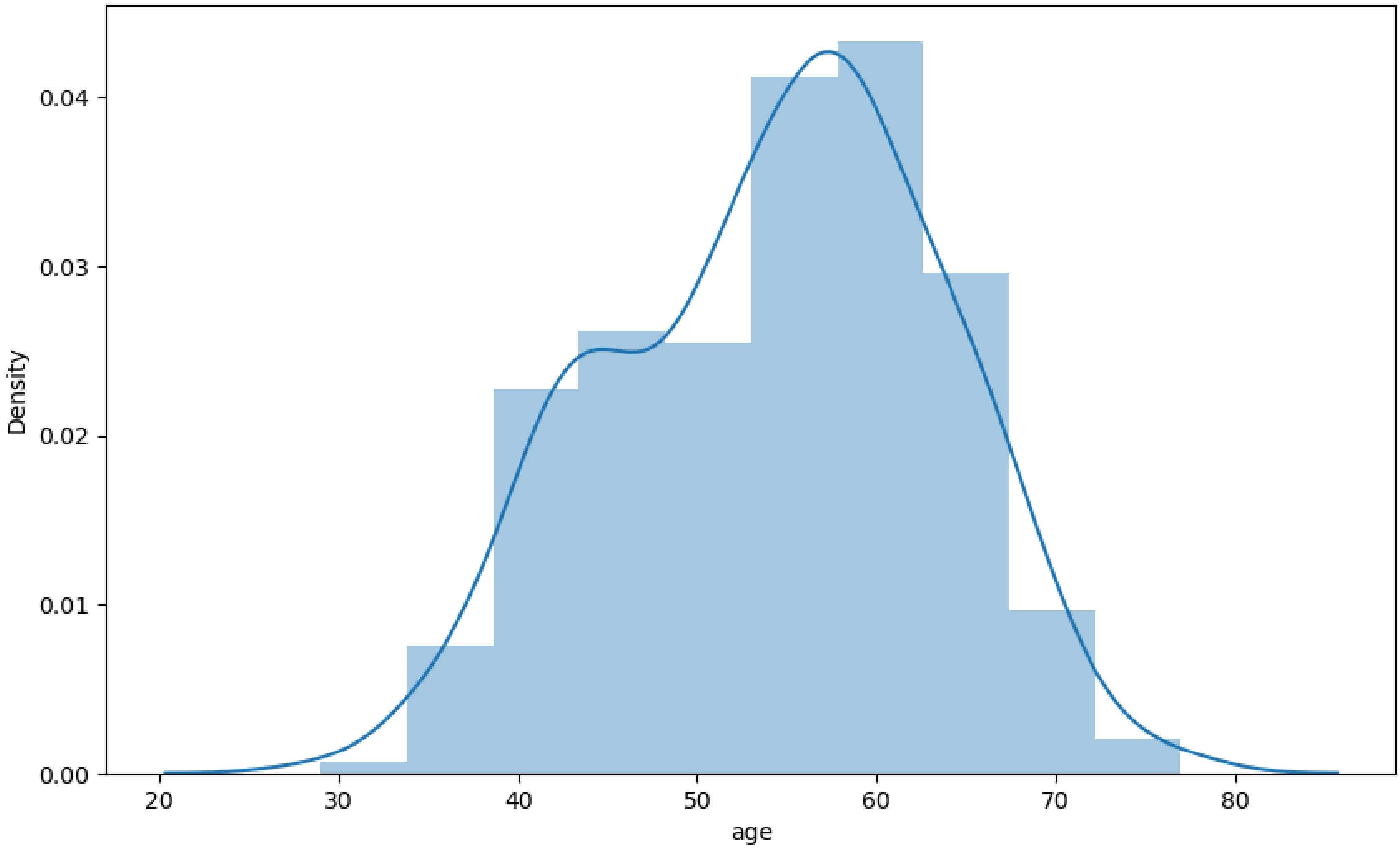
```
df['age'].describe()
```

Out[260…

```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

In [262…

```
f,ax = plt.subplots(figsize=(10,6))
x=df['age']
ax=sns.distplot(x,bins=10)
```

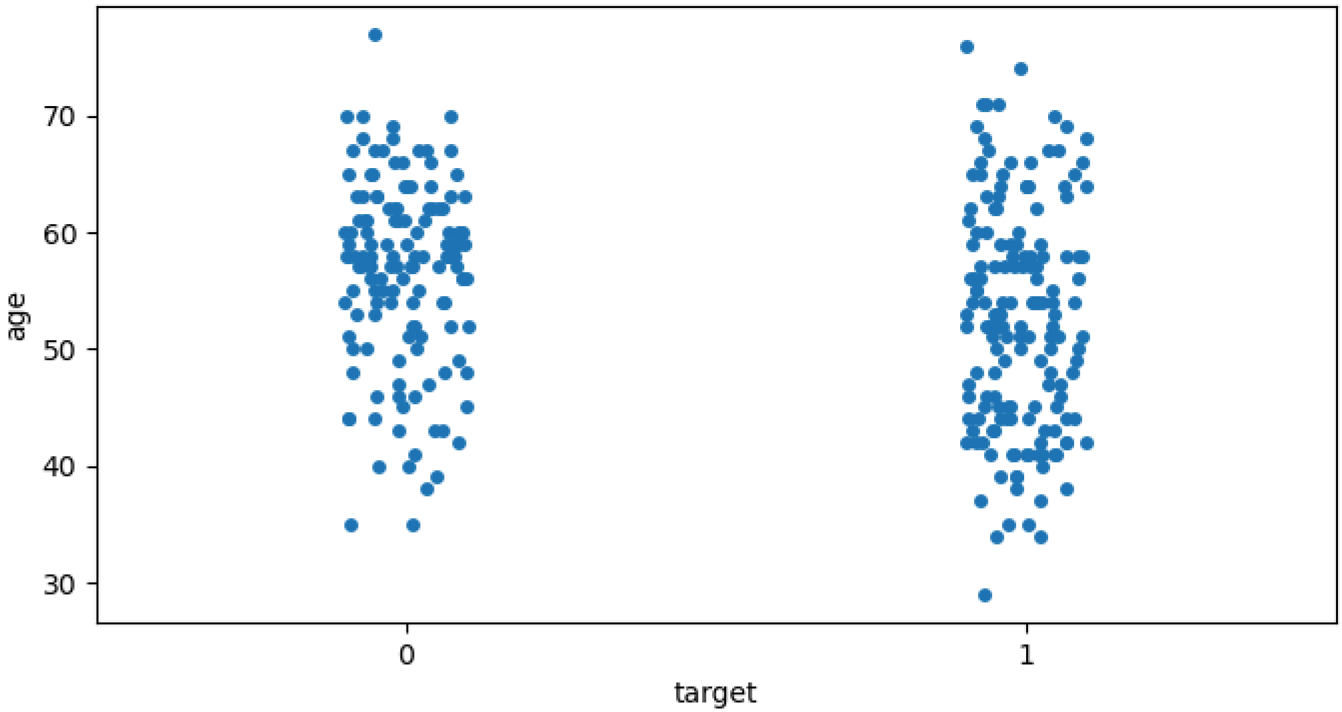


In [266…

```
f,ax= plt.subplots(figsize=(8,4))
sns.stripplot(x= 'target',y='age',data=df)
```

Out[266…

```
<Axes: xlabel='target', ylabel='age'>
```

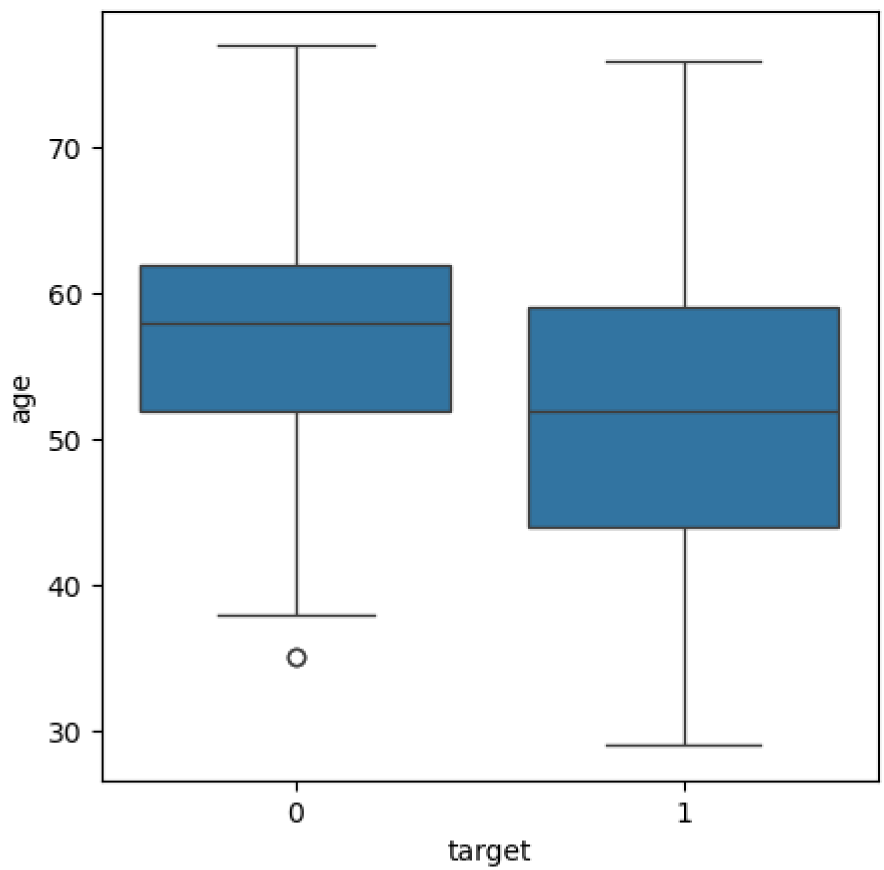


In [274…

```
f,ax = plt.subplots(figsize=(5,5))
sns.boxplot(x='target',y='age',data=df)
```

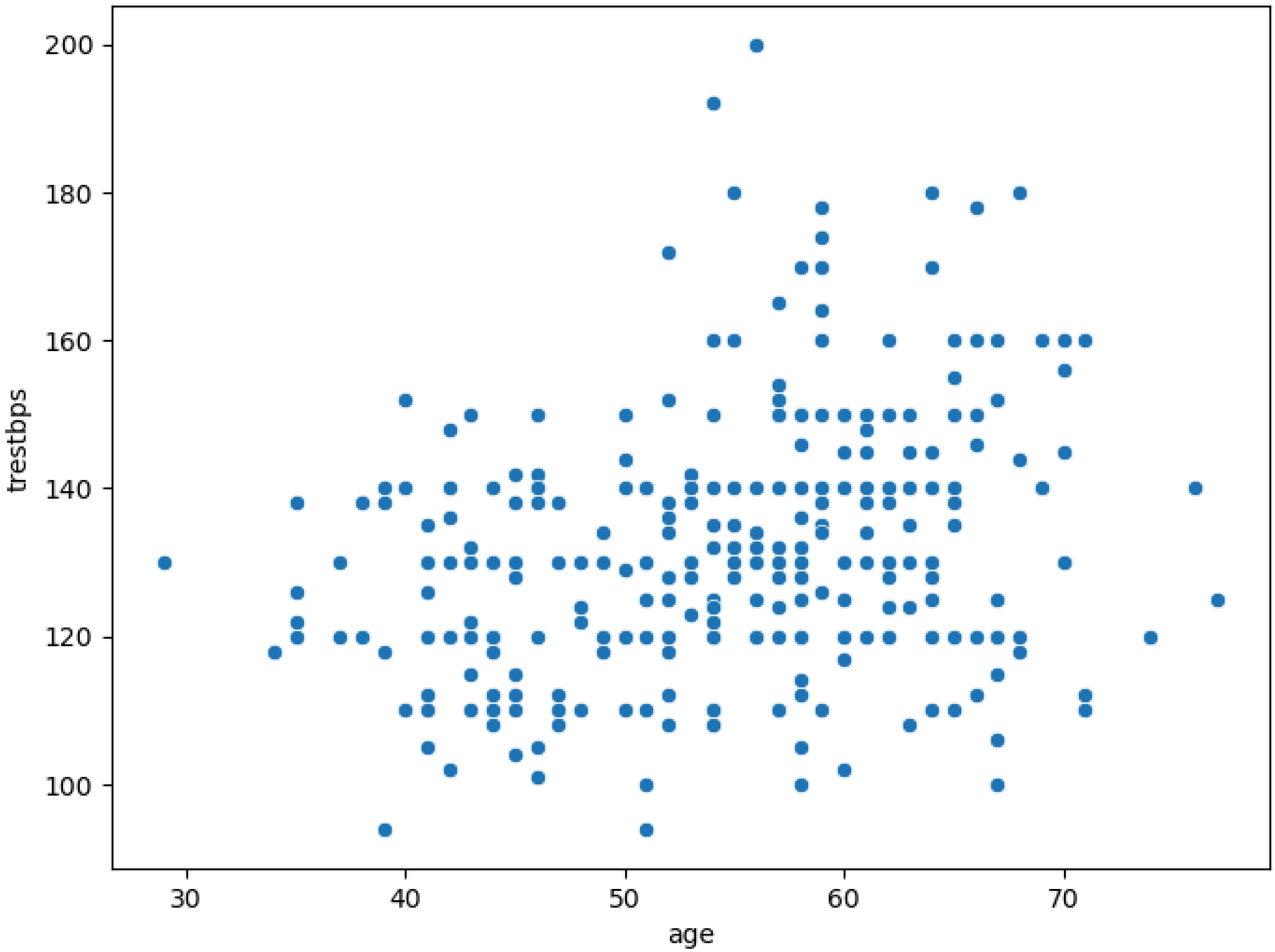
Out[274…

```
<Axes: xlabel='target', ylabel='age'>
```



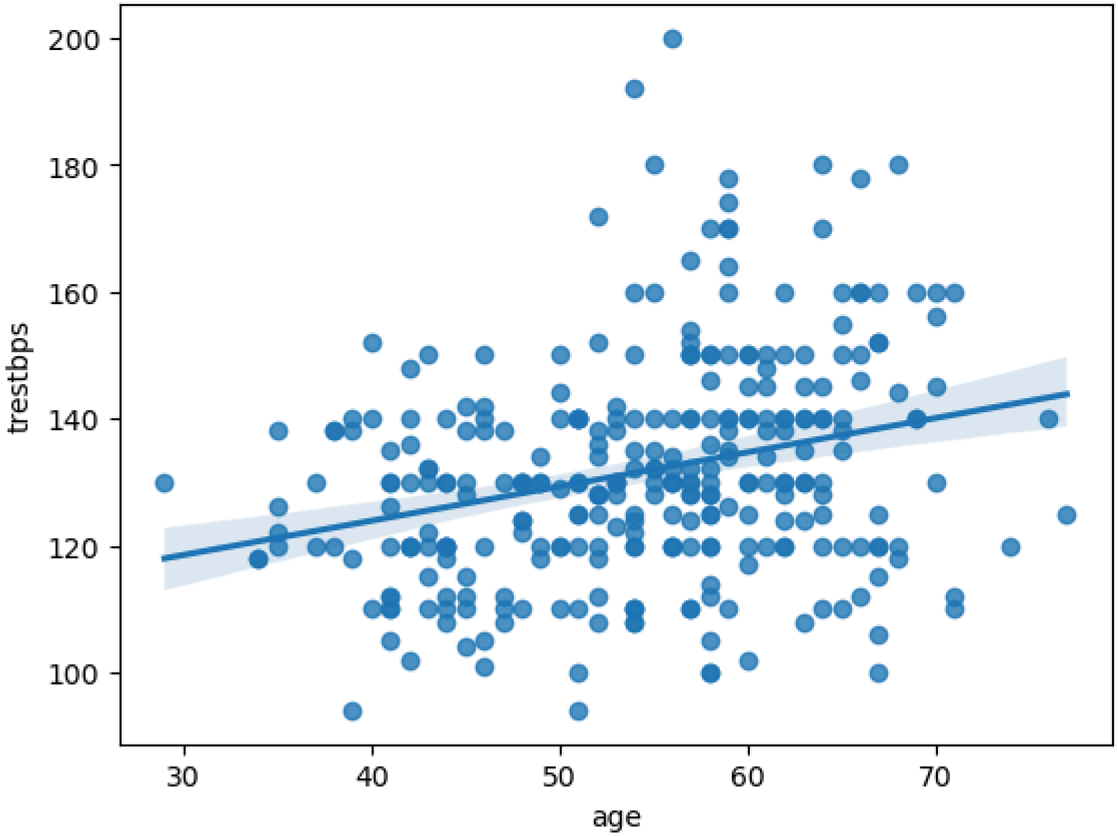
In [280…

```
f,ax = plt.subplots(figsize=(8,6))
ax =sns.scatterplot(x='age',y='trestbps',data=df)
```



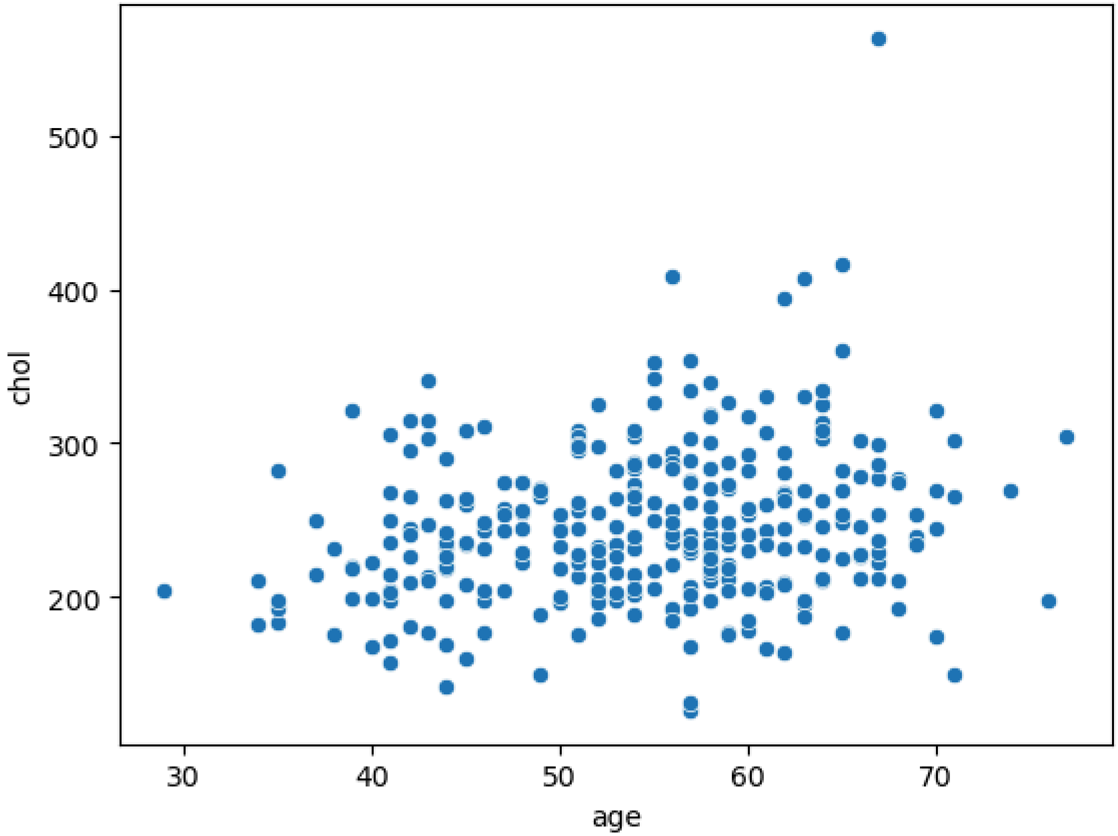
In [282…

```
ax= sns.regplot(x='age',y='trestbps',data=df)
```



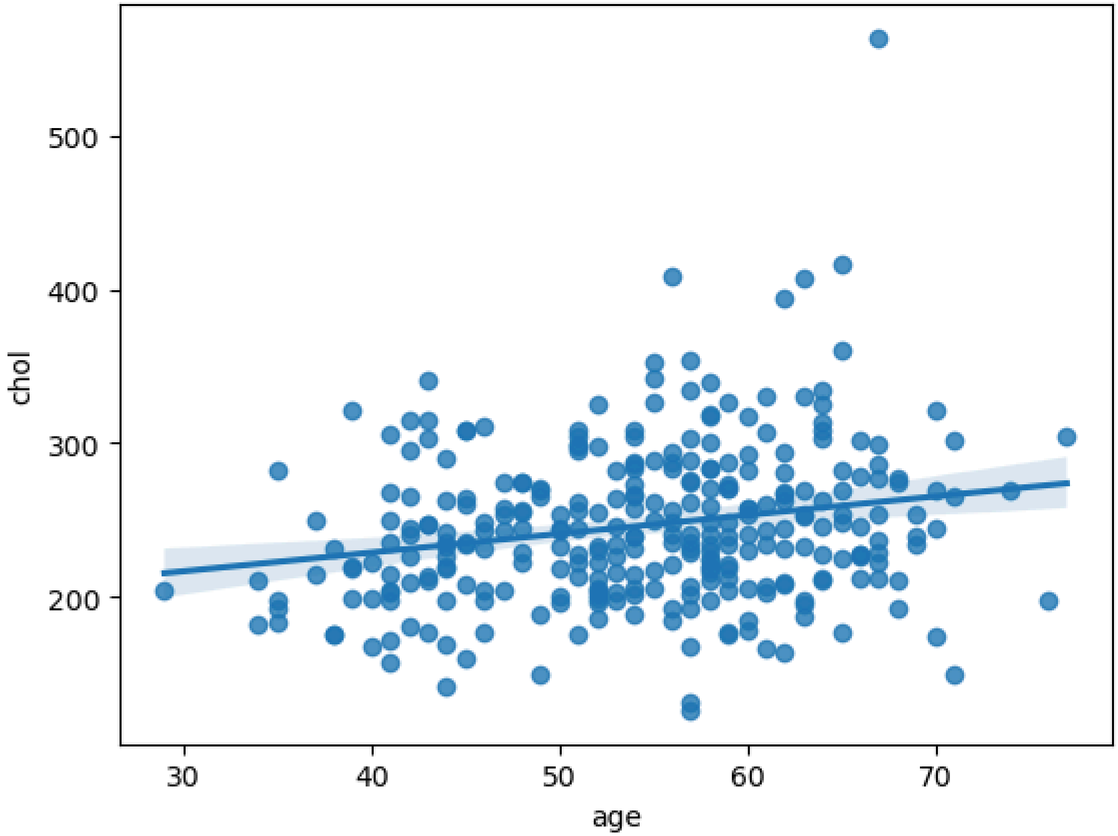
In [286…

```
ax=sns.scatterplot(x='age',y='chol',data=df)
```



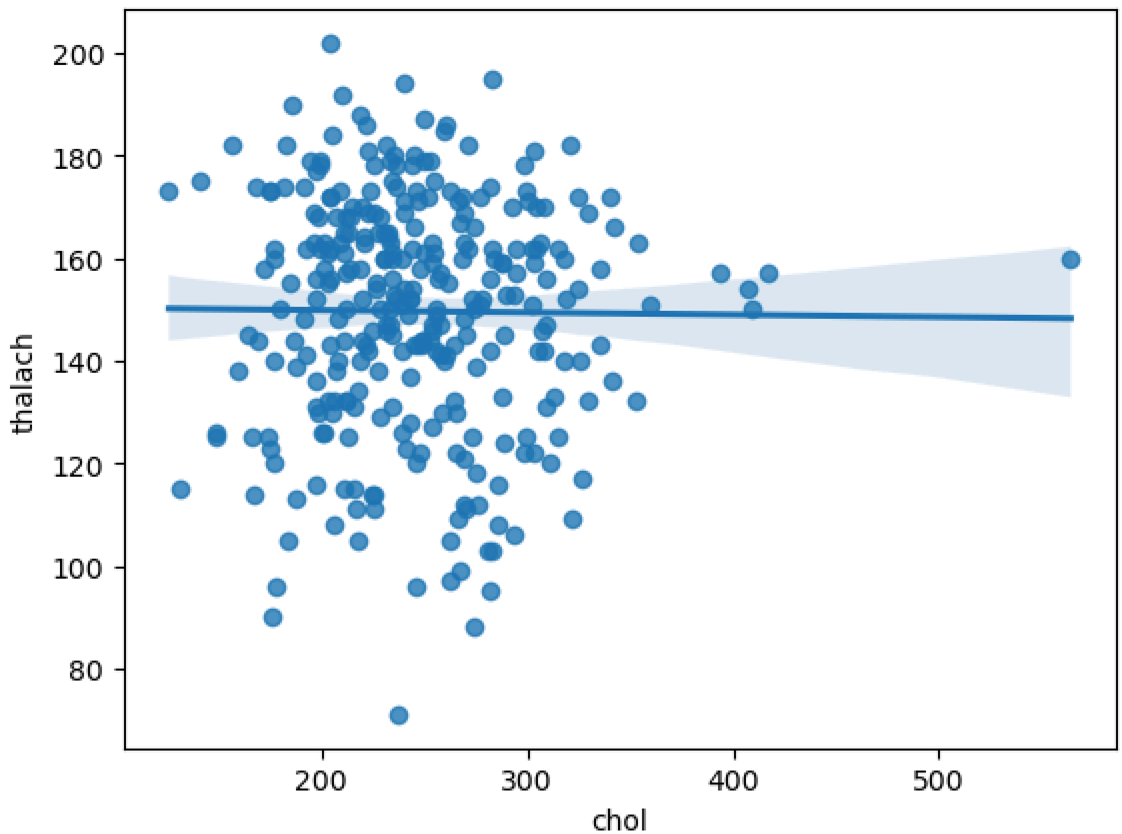
In [288…
```
sns.regplot(x='age',y='chol',data=df)
```

Out[288…
```
<Axes: xlabel='age', ylabel='chol'>
```



In [292…
```
sns.regplot(x='chol',y='thalach',data=df)
```

Out[292…
```
<Axes: xlabel='chol', ylabel='thalach'>
```



In [296…

```
df.isnull().sum()
```

Out[296…

```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

In [298…

```
df.isnull().sum().sum()
```

Out[298…

```
0
```

In [300…

```
df.notnull()
```

Out[300…

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	True	True	True	True	True	True	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True	True	True	True	True	True	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	True	True	True	True	True	True	True	True	True	True	True	True	True	True
299	True	True	True	True	True	True	True	True	True	True	True	True	True	True
300	True	True	True	True	True	True	True	True	True	True	True	True	True	True
301	True	True	True	True	True	True	True	True	True	True	True	True	True	True
302	True	True	True	True	True	True	True	True	True	True	True	True	True	True

303 rows × 14 columns

In [302…

```
df.notnull().sum()
```

Out[302…

```
age         303
sex         303
cp          303
trestbps    303
chol        303
fbs         303
restecg     303
thalach     303
exang       303
oldpeak     303
slope       303
ca          303
thal        303
target      303
dtype: int64
```

In [310…

```
assert pd.notnull(df).all().all()
```

In [320…

```
assert (df >= 0).all().all()
```

In [324…

```
df['age'].describe()
```

Out[324…

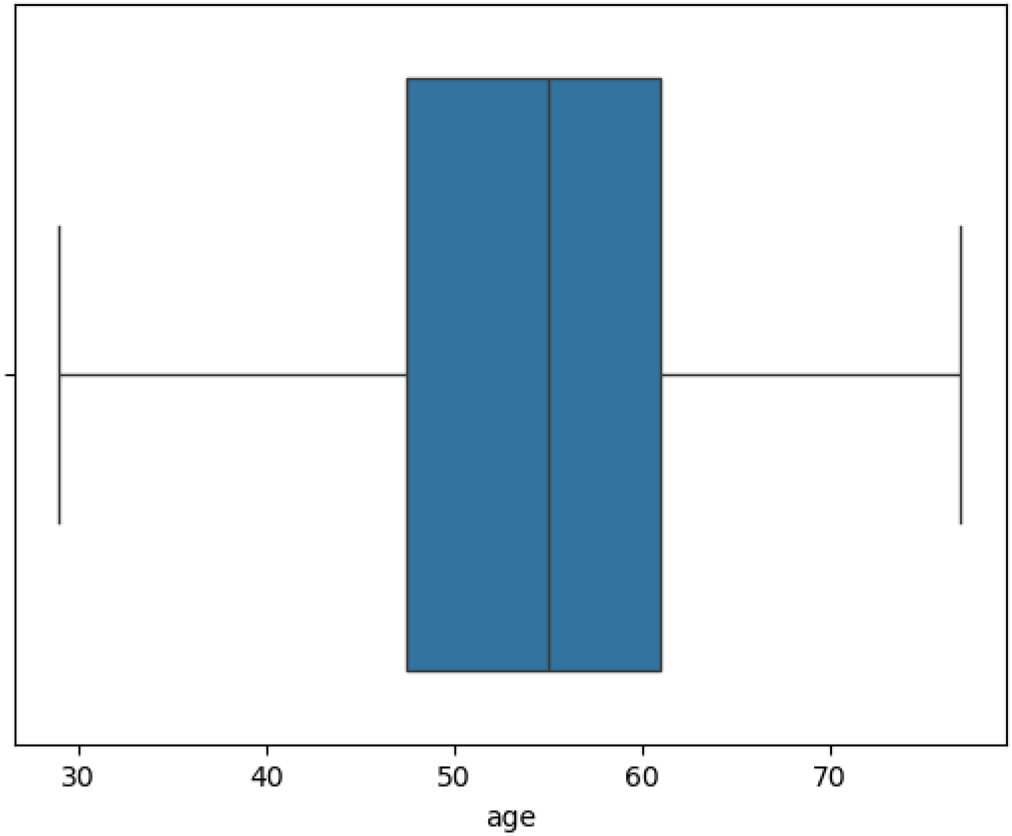
```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

In [334…

```
sns.boxplot(x=df['age'])
```

Out[334…

```
<Axes: xlabel='age'>
```

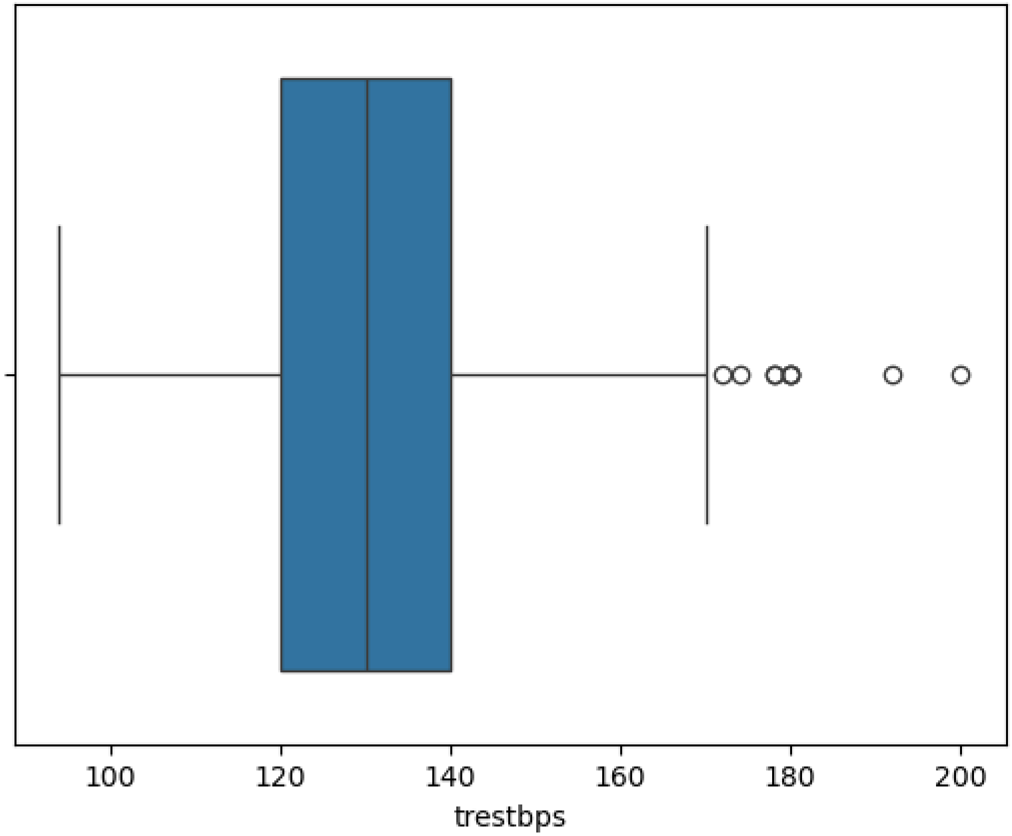


In [336…
```
df['trestbps'].describe()
```

Out[336…
```
count    303.000000
mean     131.623762
std       17.538143
min       94.000000
25%      120.000000
50%      130.000000
75%      140.000000
max      200.000000
Name: trestbps, dtype: float64
```

In [338…
```
sns.boxplot(x=df['trestbps'])
```

Out[338…
```
<Axes: xlabel='trestbps'>
```



In [340…

```
df['chol'].describe()
```

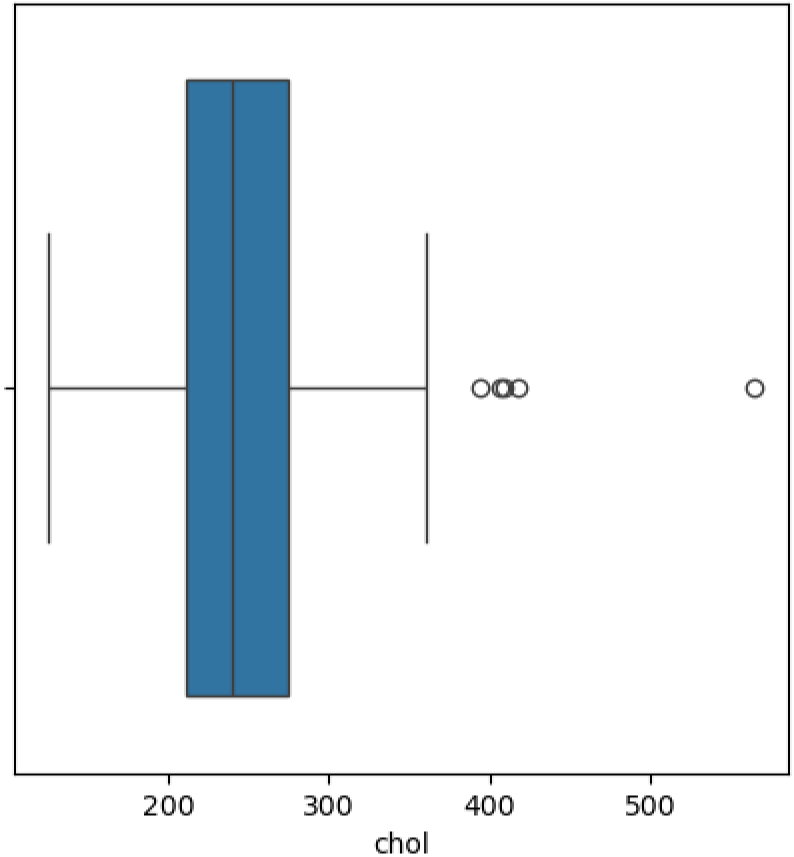
Out[340…

```
count    303.000000
mean     246.264026
std       51.830751
min      126.000000
25%      211.000000
50%      240.000000
75%      274.500000
max      564.000000
Name: chol, dtype: float64
```

In [344…

```
plt.subplots(figsize=(5,5))
sns.boxplot(x=df['chol'])
```

Out[344… <Axes: xlabel='chol'>

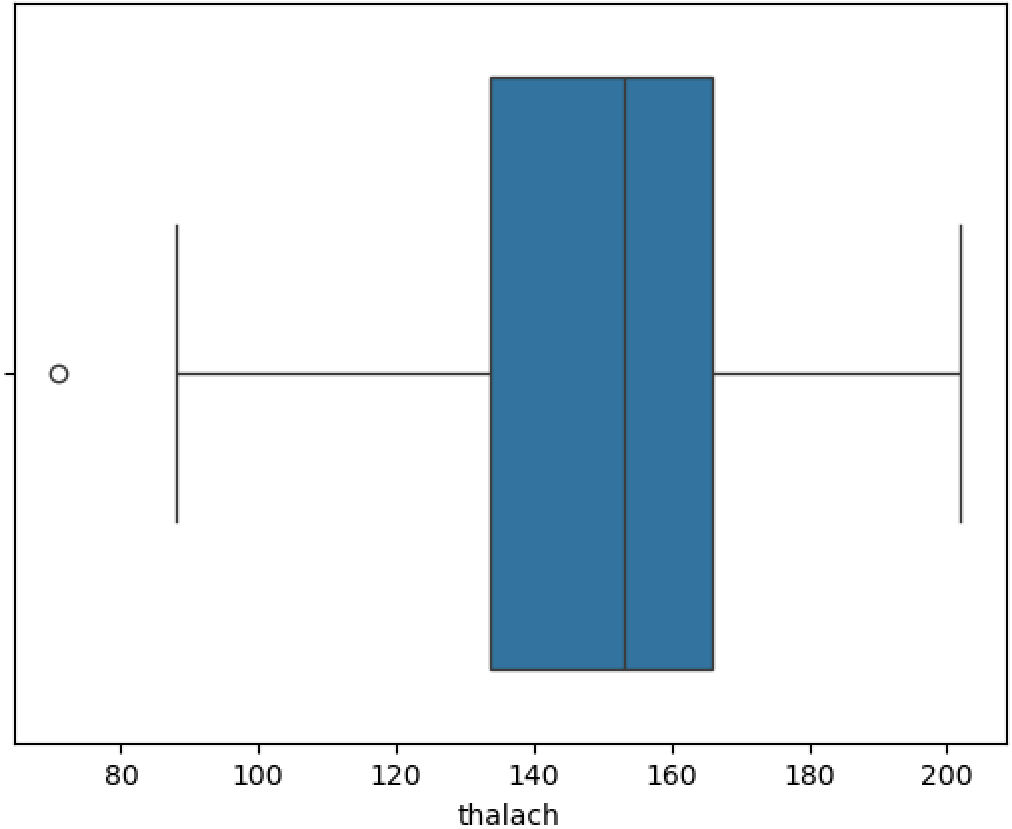


In [346…
```
df['thalach'].describe()
```

Out[346…
```
count    303.000000
mean     149.646865
std       22.905161
min       71.000000
25%      133.500000
50%      153.000000
75%      166.000000
max      202.000000
Name: thalach, dtype: float64
```

In [348…
```
sns.boxplot(x=df['thalach'])
```

Out[348…
```
<Axes: xlabel='thalach'>
```

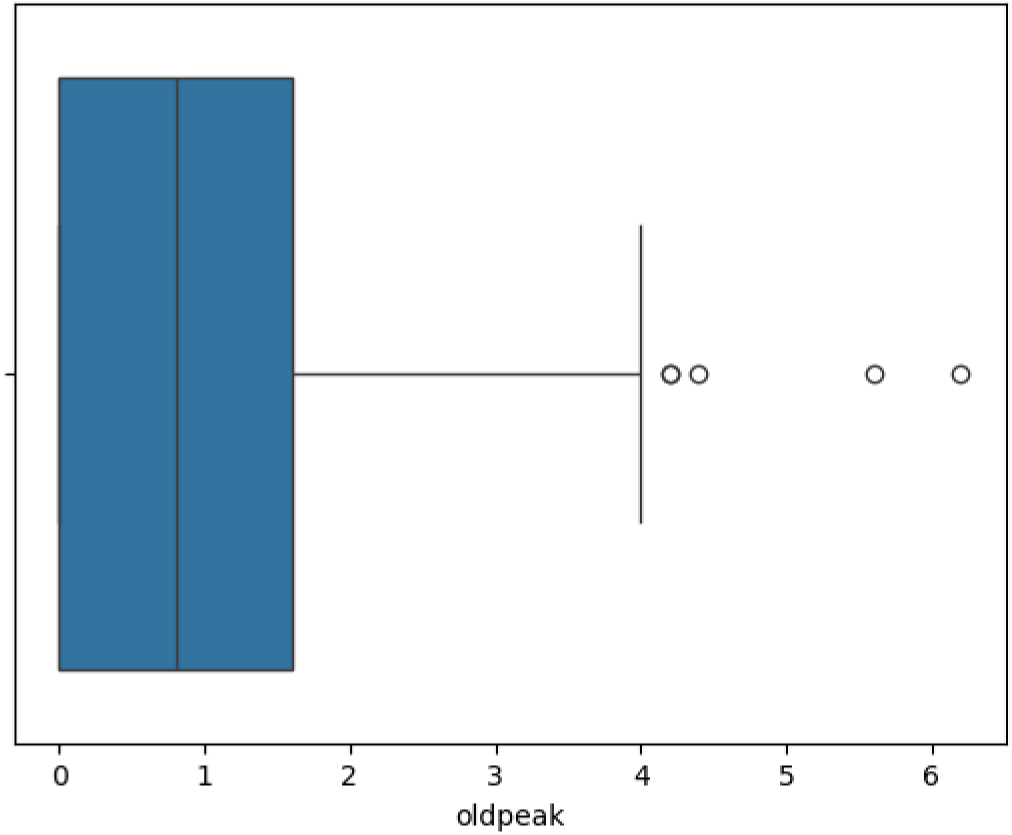


In [350…
```
df['oldpeak'].describe()
```

Out[350…
```
count    303.000000
mean       1.039604
std        1.161075
min        0.000000
25%        0.000000
50%        0.800000
75%        1.600000
max        6.200000
Name: oldpeak, dtype: float64
```

In [354...
```
sns.boxplot(x=df['oldpeak'])
```

Out[354...
```
<Axes: xlabel='oldpeak'>
```



In [356...
```
cor=df.corr()
```

In [358...
```
cor
```

Out[358…

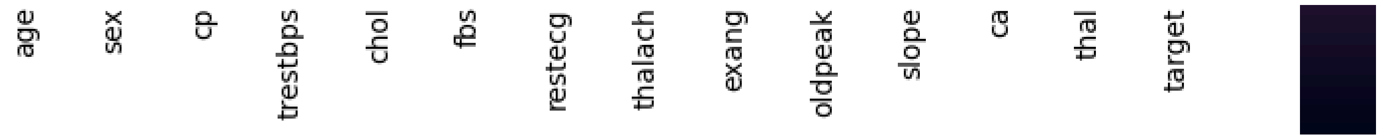
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
age	1.000000	-0.098447	-0.068653	0.279351	0.213678	0.121308	-0.116211	-0.398522	0.096801	0.210013	-0.168814
sex	-0.098447	1.000000	-0.049353	-0.056769	-0.197912	0.045032	-0.058196	-0.044020	0.141664	0.096093	-0.030711
cp	-0.068653	-0.049353	1.000000	0.047608	-0.076904	0.094444	0.044421	0.295762	-0.394280	-0.149230	0.119717
trestbps	0.279351	-0.056769	0.047608	1.000000	0.123174	0.177531	-0.114103	-0.046698	0.067616	0.193216	-0.121475
chol	0.213678	-0.197912	-0.076904	0.123174	1.000000	0.013294	-0.151040	-0.009940	0.067023	0.053952	-0.004038
fbs	0.121308	0.045032	0.094444	0.177531	0.013294	1.000000	-0.084189	-0.008567	0.025665	0.005747	-0.059894
restecg	-0.116211	-0.058196	0.044421	-0.114103	-0.151040	-0.084189	1.000000	0.044123	-0.070733	-0.058770	0.093045
thalach	-0.398522	-0.044020	0.295762	-0.046698	-0.009940	-0.008567	0.044123	1.000000	-0.378812	-0.344187	0.386784
exang	0.096801	0.141664	-0.394280	0.067616	0.067023	0.025665	-0.070733	-0.378812	1.000000	0.288223	-0.257748
oldpeak	0.210013	0.096093	-0.149230	0.193216	0.053952	0.005747	-0.058770	-0.344187	0.288223	1.000000	-0.577537
slope	-0.168814	-0.030711	0.119717	-0.121475	-0.004038	-0.059894	0.093045	0.386784	-0.257748	-0.577537	1.000000
ca	0.276326	0.118261	-0.181053	0.101389	0.070511	0.137979	-0.072042	-0.213177	0.115739	0.222682	-0.080155
thal	0.068001	0.210041	-0.161736	0.062210	0.098803	-0.032019	-0.011981	-0.096439	0.206754	0.210244	-0.104764
target	-0.225439	-0.280937	0.433798	-0.144931	-0.085239	-0.028046	0.137230	0.421741	-0.436757	-0.430696	0.345877

In [388…

```
f,ax= plt.subplots(figsize=(10,10))
corr= df.corr()
a=sns.heatmap(corr,square=True,annot=True,linecolor='white')
a.set_xticklabels(a.get_xticklabels(),rotation=90)
a.set_yticklabels(a.get_yticklabels(),rotation=30)
```

Out[388…
```
[Text(0, 0.5, 'age'),
 Text(0, 1.5, 'sex'),
 Text(0, 2.5, 'cp'),
 Text(0, 3.5, 'trestbps'),
 Text(0, 4.5, 'chol'),
 Text(0, 5.5, 'fbs'),
 Text(0, 6.5, 'restecg'),
 Text(0, 7.5, 'thalach'),
 Text(0, 8.5, 'exang'),
 Text(0, 9.5, 'oldpeak'),
 Text(0, 10.5, 'slope'),
 Text(0, 11.5, 'ca'),
 Text(0, 12.5, 'thal'),
 Text(0, 13.5, 'target')]
```


age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target

In [ ]: